

Conformational changes, excess area, and elasticity of the Piezo protein-membrane nanodome from coarse-grained and atomistic simulations

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eLife Assessment

This work represents an **important** contribution to our understanding of how membrane energetics influence protein conformation and function in mechano-sensitive channels. Through extensive molecular dynamics simulations and energetic analysis, the study **convincingly** demonstrates how the channel structure is shaped by a balance of protein and membrane-induced forces, effectively reconciling experimental data from different membrane environments. This work will appeal broadly to researchers and readers with interests in ion channel structure and function, mechanosensation, and membrane biophysics.

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Abstract

The mechanosensitive ion channels Piezo 1 and 2 induce a curved protein-membrane nanodome that flattens with increasing membrane tension γ . The tension-induced flattening of the nanodome is associated with Piezo activation and driven by the energy $\gamma\Delta A$ where ΔA is the excess area of the curved nanodome relative to its planar projected area. Based on extensive coarse-grained and atomistic simulations of membrane-embedded Piezo 1 and 2 proteins, we report here an excess area ΔA for the Piezo protein-membrane nanodome of about 40 nm^2 in tensionless membranes, and a half-maximal reduction of ΔA at tension values of about 3 to 4 mN/m, which is within the range of experimentally determined values for the half-maximal activation of Piezo 1. In line with recent experimental investigations of Piezo proteins in cell membranes and membrane vesicles, the membrane-embedded Piezo proteins adopt conformations in our simulations that are significantly less curved than the protein conformation in the detergent micelles of cryo-EM structures. An elasticity analysis of the nanodome shapes and protein conformations obtained from our simulations leads to an elastic model for Piezo activation that distinguishes the different energy components of the protein and the membrane in the tension-induced flattening of the nanodome.

Introduction

The transmembrane (TM) proteins Piezo 1 and 2 are mechanosensitive ion channels (Coste et al., 2010; 2012) that mediate numerous physiological processes in mammals, including touch sensation and blood pressure control (Wu et al., 2017). Cryo-electron microscopy (cryo-EM) structures of Piezo 1 (Guo and MacKinnon, 2017; Saotome et al., 2018; Zhao et al., 2018) and Piezo 2 (Wang et al., 2019) in detergent micelles revealed three identical monomeric arms with 38 TM helices that spiral out from the central ion channel, which is lined by the innermost TM helices of the arms. The TM domains of the helices in the three arms do not lie in a plane, which led to the suggestions that the Piezo protein curves the cell membrane into a nanodome, and that this protein-membrane nanodome flattens when external forces induce a tension γ in the membrane, leading to channel opening (Guo and MacKinnon, 2017). The flattening of the nanodome is driven by the energy $\gamma\Delta A$, where

ΔA is the excess area of the nanodome due to its curved shape, compared to the projected area of the nanodome in the plane of the surrounding membrane (Guo and MacKinnon, 2017; Haselwandter and MacKinnon, 2018). Cryo-electron tomography images of reconstituted membrane vesicles with embedded Piezo 1 proteins indeed show that the proteins induce membrane curvature (Guo and MacKinnon, 2017), but elastic modeling of vesicle shapes with varying diameters indicates that this induced curvature is on average about four times smaller than the curvature of the proteins in the detergent micelles of the high-resolution cryo-EM structures (Haselwandter et al., 2022b; Haselwandter et al., 2022a). High-resolution fluorescence imaging of Piezo 1 in its native cell membrane environment confirms an expansion in the inactivated state from flattening of the arms compared to structural models in detergent micelles, and indicates further flattening upon activation (Mul-hall et al., 2023). A cryo-EM structure of flattened Piezo 1 with widened ion channel has been obtained by embedding Piezo 1 proteins in small membrane vesicles with a diameter of 20 nm in an outside-out orientation in which the vesicle curvature opposes the intrinsic Piezo curvature (Yang et al., 2022). In patch-clamp experiments with Piezo 1 and Piezo 2, an opening of the ion channel induced by membrane tension has been observed for Piezo 1, but not for Piezo 2, which appears to indicate that Piezo 2 activation requires other factors of its cellular context that are not present in the experiments, besides membrane tension (Moroni et al., 2018).

Molecular dynamics (MD) simulations have the potential to complement recent experimental insights on the structure of Piezo proteins in their native membrane environment by providing high-resolution, dynamic information of membrane-embedded Piezo proteins under different membrane tensions (Botello-Smith et al., 2019; Chong et al., 2021; Buyan et al., 2020; Lin et al., 2022; Jiang et al., 2021; De Vecchis et al., 2021). In atomistic simulation trajectories with a length up to 100 ns at membrane tensions between 14.2 mN/m and 67.8 mN/m starting from a tensionless Piezo 1 protein-membrane nanodome with a simulation-box area of $31.4 \times 31.4 \text{ nm}^2$, De Vecchis et al. (2021) observed a flattening of the nanodome, and a significant expansion of the channel volume at the largest tension of 67.8 mN/m. In atomistic simulations of truncated Piezo 1 with TM helices 17 to 38 in a tensionless membrane, a flattening of the protein-membrane nanodome leading to channel opening has been induced by a small simulation-box area of about $480 \text{ to } 485 \text{ nm}^2$ that aims to mimic a crowding of Piezo 1 proteins (Jiang et al., 2021). The membrane flattening in these simulations is a consequence of the standard periodic boundary conditions of the simulation box, which lead to an on average vanishing slope of the membrane at the box boundaries.

In this article, we report results from coarse-grained simulations of the Piezo 1 and Piezo 2 protein-membrane nanodome with a length up to 8 μs at membrane tensions between 0 and 20.8 mN/m, starting from tensionless protein-membrane nanodomains with a simulation-box area of $50 \times$

50 nm². To corroborate these coarse-grained simulation results, we performed additional atomistic simulations of the Piezo 2 protein-membrane nanodome with a length up to 300 ns and an initial, tensionless simulation-box area of 33 × 33 nm² at membrane tensions between 0 and 18 mN/m. A main focus in our analysis is on determining the excess area ΔA of the protein-membrane nanodomains as a function of the membrane tension γ . In both our coarse-grained simulations with the Martini 2.2 force field (de Jong et al., 2013b [↗](#)) and atomistic simulations with the CHARMM36 force field (Huang and MacKerell Jr, 2013 [↗](#); Klauda et al., 2010 [↗](#)), we obtain an excess area ΔA for the protein-membrane nanodome of about 40 nm² in the tensionless state, a reduction to values of about 5 nm² and below at tension values larger than 10 mN/m at which the nanodome is nearly completely flattened, and a half-maximal response of ΔA at tension values of about 3 to 4 mN/m, which is within the range of experimentally determined values for the half-maximal activation of Piezo 1 (Lewis and Grandl, 2015 [↗](#); Cox et al., 2016 [↗](#)). In agreement with experimental observations for membrane-embedded Piezo 1 (Haselwandter et al., 2022b [↗](#),a; Mulhall et al., 2023 [↗](#)), the excess area ΔA as well as the height and curvature of the tensionless protein-membrane nanodomains is significantly reduced compared to corresponding values for the cryo-EM structure of the proteins in detergent micelles. At high membrane tensions, we observe a widening of the Piezo1 ion channel as in the flattened cryo-EM structure of Piezo 1 in small membrane vesicles (Yang et al., 2022 [↗](#)). The Piezo2 ion channel, in contrast, does not respond to membrane tension in our simulations, which is in line with patch-clamp experiments in which Piezo 1, but not Piezo 2, was shown to be activated by membrane tension alone (Moroni et al., 2018 [↗](#)). In addition to ΔA , we determine the bending energy of the lipid membrane in the nanodomains, which allows to construct an elasticity model that identifies the energetic contributions of the protein and the membrane in the response of the Piezo protein-membrane nanodome to membrane tension.

Results

Piezo 2 protein structure and nanodome shape in tensionless membranes

Our simulations of the Piezo 2 protein-membrane nanodome are based on the cryo-EM structure of the Piezo 2 trimer (PDB ID 6KG7) (Wang et al., 2019 [↗](#)), which includes all 38 TM helices of the Piezo 2 monomers. In these simulations, the Piezo 2 trimer is embedded in an asymmetric membrane that mimics essential aspects of the lipid composition of cell membranes (see Methods). We find that the relaxed simulation structure of Piezo 2 in tensionless membranes is significantly flattened compared to the cryo-EM structure of the Piezo 2 trimer in detergent micelles (see **Figure 1** [↗](#)). To compare these structures, we have performed coarse-grained simulations of membrane-embedded Piezo 2 in which the protein structure is constrained to the cryo-EM structure, in addition to simulations without such constraints in which the protein structure relaxes in the tensionless membrane within the 8 μ s of the simulation trajectories. The two representative conformations in **Figure 1(a)** [↗](#) compare the protein-membrane nanodome at the end of simulation trajectories with and without constraints on the protein structure. In these conformations, the TM helices of Piezo 2 are indicated as rods, and the shape of the membrane midplane is shown as a continuous surface determined from the positions of the lipid head beads in the two leaflets of the membranes (see Methods). The relaxed simulation structure of Piezo 2 is clearly less curved than the constrained cryo-EM structure. In addition to these different curvatures of the protein-membrane nanodome, the membrane shape in both representative conformations of **Figure 1(a)** [↗](#) exhibits undulations from thermally excited shape fluctuations.

To quantify the protein-induced curvature and excess area of the nanodome, and to eliminate the additional curvature and excess area of thermal membrane shape fluctuations present also in protein-free membranes, we average the continuous membrane-midplane shapes of individual conformations over the same time intervals of all trajectories. The average membrane shapes,

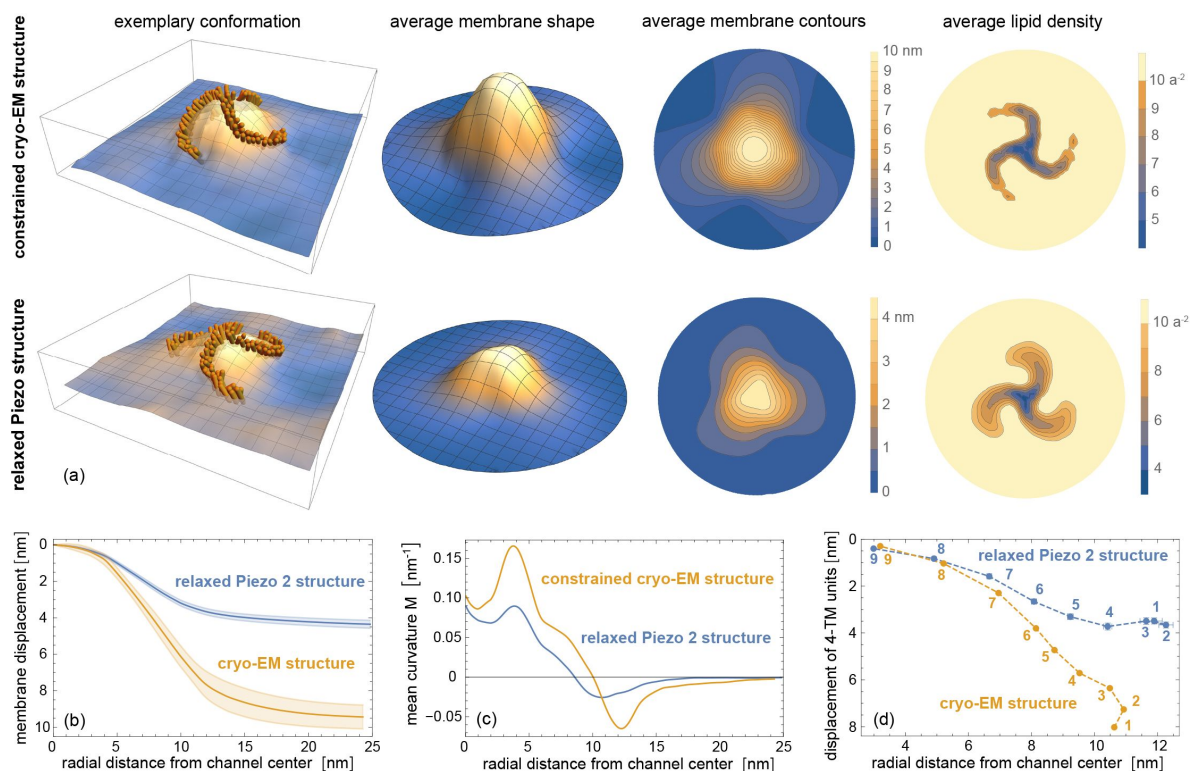


Figure 1.

Coarse-grained Piezo 2 nanodome in a tensionless membrane.

(a) Representative conformations and average nanodome shape, average nanodome contours, and average lipid density within quadratic membrane patches with side length $a \approx 2$ nm for the constrained Piezo 2 cryo-EM structure and the relaxed Piezo 2 structure in a coarse-grained tensionless membrane. The relaxed nanodome shape is an average over conformations from the last microseconds of 10 independent simulation runs with a length of 8 μ s. The nanodome shape for the constrained Piezo 2 cryo-EM structure is an average over the last microseconds of three independent simulation runs with a length of 4 μ s for a harmonic force constant of 1000 kJ mol⁻¹ nm⁻² per bead. (b) Radial membrane profiles of the average nanodome shapes. The shaded areas represent the error of the mean obtained from the shape profiles of the independent runs. (c) Mean curvature M along the shape profiles calculated from first and second derivatives at radial distance r , which are obtained from local quadratic fits of profile segments $r \pm 2$ nm. (d) Average displacement of the center of mass (COM) of 4 TM-helix units versus average radial distance. The 4 TM-helix units are numbered as in [Figure 7](#). Only protein residues in the TM sections of the 4-TM units are included in the calculation of COMs (see Methods).

membrane contours, and lipid densities in **Figure 1(a)** are averaged over the conformations in the last microseconds of all 10 simulation trajectories of length 8 μs with the relaxed Piezo 2 protein, and over the last microseconds of the 3 simulation trajectories with the constrained protein (see Methods). Because of the translational and rotational diffusion of the Piezo proteins along the simulation trajectories, the averaging of conformations requires an alignment of the protein (see Methods). The average membrane shapes and lipid densities in **Figure 1(a)** therefore have a circular projected area with a diameter that corresponds to the average x and y dimensions of the simulation boxes of the conformations, because only these ‘inscribed’ circular membrane segments are overlying in all rotationally aligned conformations. From the simulation trajectories in which the protein structure is constrained to the cryo-EM structure of Piezo 2 in detergent micelles, we obtain a nanodome midplane shape with a height of 10 nm, which is significantly larger than the height of about 4 nm for the nanodome shape with the relaxed Piezo 2 protein. The mean curvature profiles of the nanodome shapes in **Figure 1(c)**, which we calculate from the radially averaged nanodome profiles of **Figure 1(b)**, exhibit maxima at a radial distance of about 4 nm from the channel. In these mean curvature profiles, the maximal mean curvature of the nanodome with constrained protein structure is about two times larger than the maximal mean curvature of the nanodome with the relaxed protein. The mean curvature of the nanodome with the relaxed Piezo 2 protein drops to 0 for radial distances larger than about 17 nm from the channel center, which indicates a catenoidal shape of the non-planar surrounding membrane with a principal curvature in the radial direction of the shape profile in **Figure 1(b)** that is oppositely equal to the principal curvature in the perpendicular, tangential direction (Haselwandter and MacKinnon, 2018).

Tension-induced flattening of the protein-membrane nanodome

With increasing membrane tension, the nanodome shape and coarse-grained simulation structure of the membrane-embedded Piezo 2 protein flattens further (see **Figure 2**). The average height of the nanodome decreases from 4.4 nm in tensionless membranes to about 3.5 nm, 2.9 nm, and 2.1 nm at the intermediate membrane tensions γ of 1.4 mN/m, 2.8 mN/m, and 5.5 mN/m of our simulations, and to 1.6 nm and 0.9 nm at the largest tensions of 10.8 mN/m and 20.8 mN/m (see radial nanodome profiles in **Figure 2(d)**). In atomistic simulations of membrane-embedded Piezo 2, we observe a comparable tension-induced flattening of the Piezo 2 protein-membrane nanodome, and a comparable height of the nanodome in tensionless membranes, which corroborates our coarse-grained simulation results (see **Figure 3**).

To obtain values of the excess area ΔA that are not limited by the finite simulation time, in particular for the atomistic simulations, we divide all simulation trajectories into 5 equal intervals and determine the nanodome shape in each interval by averaging over the conformations of all independent simulation runs in this interval. In **Figure 4(a)** to **(c)**, we plot the excess area ΔA of the nanodome shape in the last time intervals versus the inverse time t^{-1} of the interval midpoints, for our coarse-grained simulations of membrane-embedded Piezo 1 and Piezo 2 proteins and our atomistic simulations of the membrane-embedded Piezo 2 protein. The excess area ΔA here was calculated as the area difference between the curved nanodome shape and the planar projection of the nanodome (see Methods). As starting points of our coarse-grained simulations of membrane-embedded Piezo 1, we devised a structure of full-length Piezo 1 by adding the unresolved 12 N-terminal TM helices to the Piezo 1 cryo-EM structure with PDB ID 6B3R (Guo and MacKinnon, 2017) via homology modelling (see Methods). In all simulation systems, we observe a decrease of ΔA with increasing simulation time, and therefore extrapolate the ΔA values to $t^{-1} = 0$ by linear fitting (dashed lines and shaded prediction error bands). In **Figure 3(e)**, the extrapolated ΔA values are plotted versus the membrane tension γ in the simulations. In all simulation systems, we obtain a maximal excess area ΔA for the protein-membrane nanodome of about 40 nm² in tensionless membrane, and half-maximal ΔA values at tensions of about 3 to 4 mN/m, which is within the range of experimentally determined values for the half-maximal activation of Piezo 1 (Lewis and Grandl, 2015; Cox et al., 2016). The

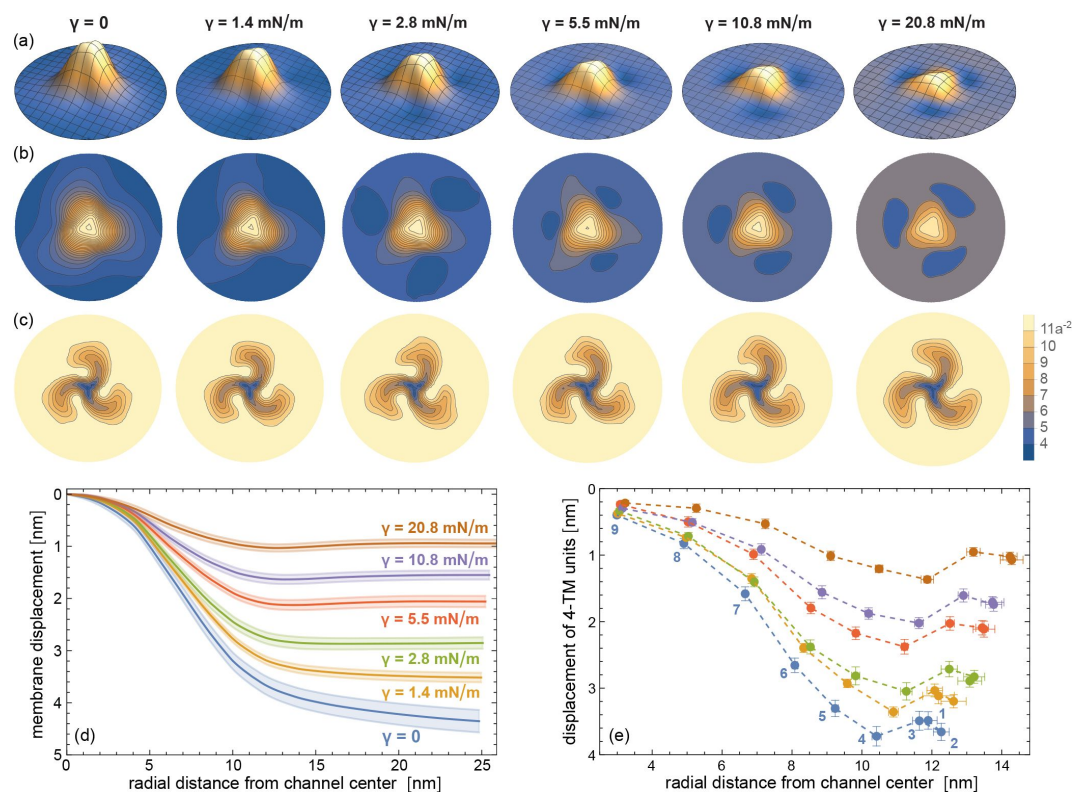


Figure 2.

Tension-induced flattening of the coarse-grained Piezo 2 protein-membrane nanodome.

(a) Average nanodome shapes, (b) average nanodome contours, and (c) average lipid density within quadratic membrane patches with sidelength $a \approx 2$ nm at membrane tensions γ from 0 to 20.8 mN/m. The nanodome shapes and lipid densities are averages over conformations from the last microseconds of 10 independent simulation runs with lengths of 8 μ s for $\gamma = 0$ and with lengths of 4 μ s for $\gamma > 0$. (d) Radial membrane profiles of the average nanodome shapes in (a). The shaded areas represent the error of the mean obtained from the shape profiles of the independent runs. (e) Average displacement of the center of mass (COM) of 4 TM-helix units versus average radial distance. The 4 TM-helix units are numbered as in [Figure 7](#). Only protein residues in the TM sections of the 4-TM units are included in the calculation of COMs.

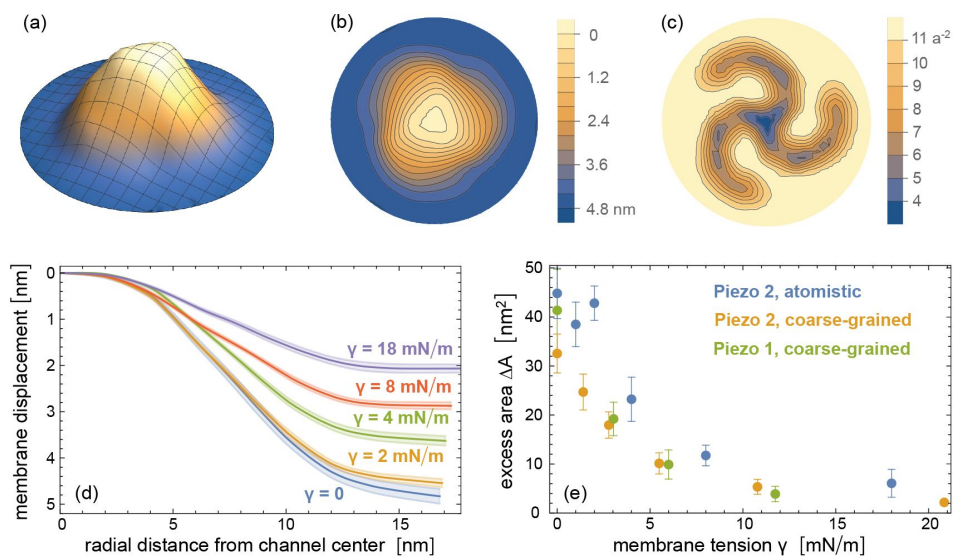


Figure 3.

Flattening of the Piezo 2 protein-membrane nanodome in atomistic simulations.

(a) Average nanodome shape, (b) average nanodome contours, and (c) average lipid density in a tensionless membrane. The nanodome shapes and lipid densities are averages over conformations from the last 50 ns of 5 independent atomistic simulation runs with lengths of 300 ns. (d) Radial membrane profiles of the nanodome shapes at membrane tensions γ from 0 to 18 mN/m obtained from averaging over conformations from the last 50 ns of 5 independent atomistic simulation runs. (e) Excess area ΔA of the protein-membrane nanodome versus membrane tension γ from coarse-grained simulations of membrane-embedded Piezo 1 and Piezo 2 and atomistic simulations with Piezo 2. The values and errors of excess area ΔA are obtained from the extrapolations to long timescales shown in [Fig. 4](#).

nanodome excess area ΔA of about 40 nm^2 in tensionless membranes is significantly smaller than the excess area $\Delta A = 165 \pm 1 \text{ nm}^2$ of the average membrane nanodome shape with constrained cryo-EM structure of Piezo 2 shown in **Figure 1(a)**.

Elasticity model of the Piezo protein-membrane nanodome

We now aim to construct a model for the elastic energies of the Piezo protein and the membrane in the tension-induced nanodome flattening observed in our simulations. The nanodome height in tensionless membranes and the tension-induced flattening of the nanodome are a consequence of opposing protein and membrane energies. The energy of the membrane is the sum $E_m = E_b + \gamma \Delta A$ of the membrane bending energy E_b and the tension-associated energy $\gamma \Delta A$, and ‘prefers’ a planar state in which this energy tends to zero. The energy E_p of the Piezo protein conformation, in contrast, prefers a curved nanodome state. The nanodome height is determined by the sum of these energies (Guo and MacKinnon, 2017; Haselwandter and MacKinnon, 2018), i.e. by the total energy

$$E_{\text{tot}} = E_p + E_m = E_p + E_b + \gamma \Delta A \quad (1)$$

To describe the tension-dependent Piezo protein conformation, we first note that the vertical displacement z_4 of the fourth 4-TM unit of the Piezo 2 arms relative to the channel center is larger than the vertical displacements z_i of the other 4-TM units in **Figure 2(e)**. We therefore use the vertical displacement z_4 of 4-TM unit 4 to describe the vertical height of the Piezo 2 TM domain, and the shift $\Delta z(\gamma) = z_4(0) - z_4(\gamma)$ of this displacement in comparison to the tensionless nanodome as parameter for the tension-induced flattening of Piezo 2. **Figure 5(a)** illustrates that the Piezo 2 height change Δz in our coarse-grained simulations is proportional to the membrane tension γ for tension values up to 6 mN/m .

At the equilibrated Piezo 2 height obtained at a given tension value, the vertical forces resulting from the protein elasticity and from the membrane elasticity are oppositely equal. If this was not the case, the Piezo 2 height would further increase or decrease until such a force balance is achieved at this tension value. We can therefore calculate the vertical force exerted by the flattening-resisting protein from the opposing force exerted by the membrane. At a given membrane tension γ , the vertical force resulting from the membrane energy E_m is

$$F = \left. \frac{dE_m}{d\Delta z} \right|_{\gamma=\text{const.}} = \frac{dE_b}{d\Delta z} + \gamma \frac{d\Delta A}{d\Delta z} \quad (2)$$

This force is the sum of two terms, and calculating these terms requires to determine the membrane bending energy E_b and the excess area ΔA of the nanodome as functions of the Piezo height change Δz . In **Figure 5(b)**, we plot the ΔA values obtained from the coarse-grained simulations of membrane-embedded Piezo 2 at different tensions versus the Δz values of the protein at these tensions, which indicates a linear relation within the error margins obtained from the simulations. From the linear fit in **Figure 5(b)**, we obtain $d\Delta A/d\Delta z = -14.6 \pm 1.8 \text{ nm}$.

For calculating the membrane bending energy of the averaged nanodome shapes, we use a discretized version of the bending energy and include the lipid densities shown in **Figure 2(c)** in order to limit the bending energy calculations to the lipid membrane of the nanodome (see Methods). To avoid limitations due to the finite simulation times, we extrapolate the bending energy values obtained for the average nanodome shapes in 5 intervals of the simulation trajectories to long timescales akin to the temporal extrapolations of the excess area ΔA (see **Figure 4(d)** and **(e)**).

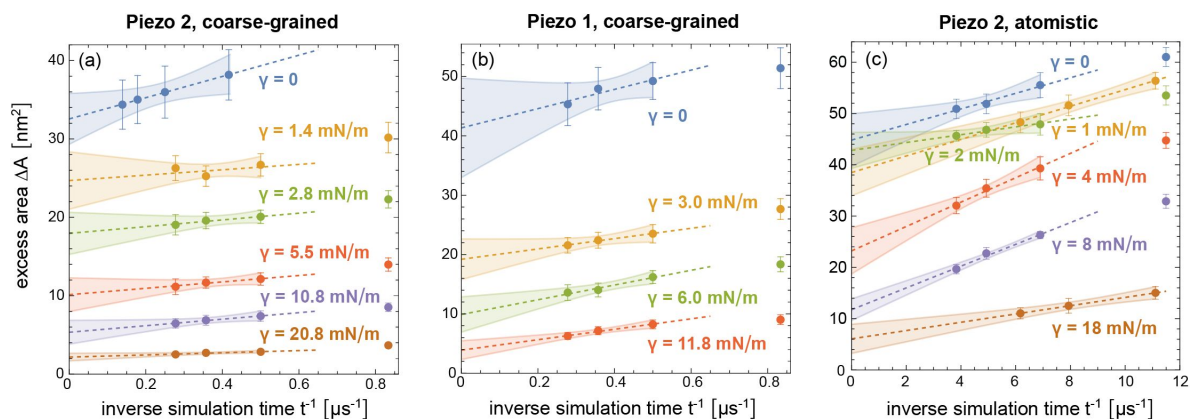


Figure 4.

Extrapolation of the excess area ΔA of the protein-membrane nanodome in coarse-grained simulations with (a) Piezo 2 and (b) Piezo 1 and (c) in atomistic simulations with Piezo 2 to long timescales.

In these extrapolations, the trajectories lengths are divided into 5 equal time intervals, and the excess area ΔA in each time interval is calculated for the nanodome shape obtained from averaging over conformations of all independent simulation runs in this time interval. The ΔA values are plotted against the inverse of the centers of the time intervals, and the values of the last three intervals are linearly fitted with the function `LinearModelFit` of Mathematica 13. The errors of the data points represent the error of the mean of ΔA values obtained for the individual simulation runs, and the shaded error region of the linear fits represent prediction bands with confidence level 0.5. The ΔA values with error shown in [Figure 4](#) are determined as the extrapolated values with prediction band error at $t^{-1} = 0$.

Figure 4—figure supplement 1. Extrapolation of the membrane bending energy E_b of the nanodome in coarse-grained simulations with Piezo 1 and 2 to long timescales.

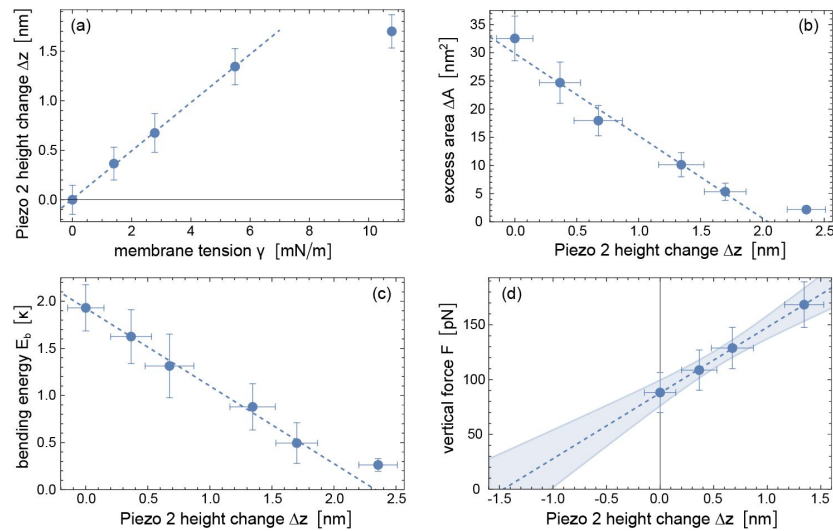


Figure 5.

Elasticity modelling of the Piezo 2 protein based on coarse-grained simulation data.

(a) Piezo 2 height change Δz versus membrane tension γ in our coarse-grained simulations of membrane-embedded Piezo 2. The protein height change is defined as $\Delta z(\gamma) = z_4(0) - z_4(\gamma)$ where z_4 is the vertical displacement of the fourth 4-TM unit of the Piezo arms relative to the channel center (see **Figure 2(e)**). (b) Excess area ΔA of the nanodome and (c) bending energy E_b of the lipid membrane in the nanodome versus Piezo height change Δz . The values and errors of ΔA and E_b are obtained from extrapolating simulation results at the tension values $\gamma = 0, 1.4, 2.8, 5.5, 10.8$, and 20.8 mN/m to long timescales (see **Figure 4(a)** and **Figure 4—figure supplement 1 (a)**). (d) Vertical force F determined from **Equation (2)** and the slopes of the fitted lines in (b) and (c) at the Δz values obtained from simulations at the tension values $\gamma = 0, 1.4, 2.8$, and 5.5 mN/m. The vertical force F is the absolute value of the opposing and in equilibrium equal forces exerted by the Piezo protein and by the membrane at a given tension value (see text). The dashed lines result from linear fitting of data points for $\gamma \leq 6$ mN/m in (a), of data points for $\Delta z < 2$ nm in (b) and (c), and of all data points in (d) with the function `LinearModelFit` of Mathematica 13. The shaded error region of the linear fit in (d) represents prediction bands with confidence level 0.5.

Figure 5—figure supplement 1. Elasticity modelling of the Piezo 1 protein based on coarse-grained simulation data.

The resulting, extrapolated bending energy values E_b of the lipid membrane in the Piezo 2 protein-membrane nanodome decrease linearly with the Piezo 2 height change Δz (see **Figure 5(c)**). From linear fitting, we obtain $d\Delta E_b/d\Delta z = -0.82 \pm 0.17 \text{ } \kappa/\text{nm}$ where κ is the bending rigidity of the lipid membrane. For the coarse-grained membrane of the lipid composition used in our simulations with main component POPC, we estimate a bending rigidity value of about $\kappa = 25 k_B T$ based on rigidity values obtained from coarse-grained simulations with the same force field Martini 2.2 for membranes composed of POPC and of various lipid mixtures (Fowler et al., 2016; Pöhnl et al., 2023; Hossein et al., 2024).

From equation 2 and the derivatives $d\Delta A/d\Delta z$ and $d\Delta E_b/d\Delta z$ determined above via linear modelling, we obtain the four data points in **Figure 5(d)** for the four Δz values at the membrane tensions $\gamma = 0, 1.4, 2.8$, and 5.5 mN/m , which indicate a linear increase of the force F with the Piezo height change Δz . From linear fitting, we obtain the force $F (\Delta z = 0) = 88 \pm 15 \text{ pN}$ for the tensionless membrane and the force increase $dF/d\Delta z = 60 \pm 20 \text{ pN/nm}$. This linear force relation is consistent with a harmonic-spring model for the elasticity of the Piezo protein. Extrapolation to zero force in **Figure 5(d)** indicates a force-free state of Piezo 2 for $\Delta z = -1.5 \pm 0.6 \text{ nm}$, which is – within the statistical accuracy – in accordance with the difference of $-2.0 \pm 0.2 \text{ nm}$ between the vertical displacement $z_4(\gamma = 0) = 3.7 \pm 0.2 \text{ nm}$ of the membrane-embedded Piezo 2 protein in tensionless membranes and the vertical displacement $z_4 = 5.7 \text{ nm}$ in the cryo-EM structure of Piezo 2 in detergent micelles (see **Figure 1(d)**). Our force modelling thus is consistent with the Piezo cryo-EM structure in detergent micelles as force-free conformation of the protein. In this modelling, the term $F (\Delta z = 0)$ depends linearly on the estimated bending rigidity value, whereas the force increase $dF/d\Delta z$ is independent of κ . For a smaller bending rigidity value of $\kappa = 20 k_B T$ instead of $25 k_B T$, we obtain $F (\Delta z = 0) = 70 \pm 12 \text{ pN}$ and a force-free state of Piezo 2 for $\Delta z = -1.2 \pm 0.5 \text{ nm}$. Our modelling assumes that any spontaneous curvature from asymmetries in the lipid composition is small compared to the curvature of the nanodome and, thus, negligible, which is plausible for the rather slight lipid asymmetry of our simulated membranes (see Methods).

From our coarse-grained simulation data of membrane-embedded Piezo 1, we obtain a linear force relation with force $F (\Delta z = 0) = 113 \pm 28 \text{ pN}$ in the tensionless state and a force increase $dF/d\Delta z = 63 \pm 30 \text{ pN/nm}$ for $\kappa = 25 k_B T$, which agrees with our results for Piezo 2 within errors (see **Figure 5-figure supplement 1**). The larger errors in the force relation for Piezo 1 compared to Piezo 2 are also a consequence of fewer tension values considered in our coarse-grained simulations of membrane-embedded Piezo 1 and, thus, fewer data points in the modelling.

Response of the Piezo 1 and 2 ion channel to large membrane tensions

Curved and flattened cryo-EM structures obtained for Piezo 1 embedded in small membrane vesicles with outside-in and outside-out protein orientation indicate a widening of the Piezo 1 ion channel in response to flattening (Yang et al., 2022). In the outside-in orientation, the extracellular side of Piezo 1 points towards the inside of the vesicles with mean diameter of 20 nm , which results in highly curved Piezo structure. In the outside-out orientation, the extracellular side of Piezo 1 points towards the vesicle outside, and the Piezo protein structure is nearly completely flattened because the vesicle curvature opposes the intrinsic Piezo curvature in this orientation. In the flattened Piezo 1 structure, the distance between the residues L2469 of the three TM helices 38 that line the ion channel is increased by 5 Å compared to the curved Piezo 1 structure, which indicates a widening of the outermost constriction site of the Piezo 1 ion channel formed by these residues (Yang et al., 2022).

In our coarse-grained simulations of membrane-embedded Piezo 1, we observe a comparable widening of the ion channel lined by the three TM helices 38 in response to tension-induced flattening of the Piezo 1 protein. At the largest simulated membrane tension $\gamma = 32 \text{ mN/m}$, the Piezo 1 protein is nearly completely flattened, and distance between the channel-constricting

residues L2469 of the three TM helices 38 is increased by about $3 \pm 1 \text{ \AA}$ compared to the Piezo 1 channel in our tensionless membranes (see **Figure 6**). This increased distance between the residues L2469 at large membrane tensions is associated with an increased tilt angle of the TM helices 38 relative to the vertical z direction of our simulation conformations. In our coarse-grained simulations of membrane-embedded Piezo 2, in contrast, we do not observe a response of the ion channel to tension (see **Figure 6**), apparently in line with patch-clamp experiments of Piezo 1 and Piezo 2, in which an tension-induced opening of the ion channel has only been observed for Piezo 1, but not for Piezo 2 (Moroni et al., 2018).

Discussion and Conclusions

Based on coarse-grained and atomistic simulations of membrane-embedded Piezo proteins, we have investigated the shape and excess area ΔA of the protein-membrane nanodome as well as the conformation of the Piezo protein in tensionless membranes and for a physiologically relevant range of membrane tensions. Our coarse-grained simulations of membrane-embedded full-length Piezo 1 and 2 proteins allowed to consider relatively large segments of about $50 \times 50 \text{ nm}^2$ on simulation timescales up to $8 \text{ }\mu\text{s}$. To assess and corroborate the validity of these coarse-grained simulations, we performed atomistic simulations of membrane-embedded full-length Piezo 2 in smaller membrane segments of about $33 \times 33 \text{ nm}^2$ on simulation timescales up to 300 ns . To quantify the protein-induced curvature and excess area of the nanodome, and to eliminate the additional curvature and excess area of thermal membrane shape fluctuations, we have determined average nanodome shapes from the simulation conformations (see **Figure 1**). We have determined the excess area ΔA from extrapolation of ΔA values for average nanodome shapes in different time intervals of the trajectories to reduce sampling errors from the finite length of the simulation trajectories (see **Figure 4**).

As a main result, we obtained an excess area ΔA of the Piezo protein-membrane nanodome of about 40 nm^2 in tensionless membranes from both coarse-grained and atomistic simulations. This excess area of the relaxed protein-membrane nanodome in tensionless membranes is significantly smaller than the excess area $\Delta A = 165 \pm 1 \text{ nm}^2$ obtained from our simulations with the constrained cryo-EM structure of Piezo 2, which in turn is comparable in magnitude to the estimate $\Delta A = 250 \text{ nm}^2$ from an analysis of the Piezo 2 dimensions in the cryo-EM structure (Wang et al., 2019). The excess area ΔA is reduced to values of about 5 nm^2 at tension values larger than 10 mN/m at which the nanodome is nearly completely flattened in our coarse-grained and atomistic simulations, and is half-maximal at tension values of about 3 to 4 mN/m , which are within the range of experimentally determined values for the half-maximal activation of Piezo 1 (Lewis and Grandl, 2015; Cox et al., 2016). Based on our results, the change of the nanodome excess area ΔA required for Piezo activation thus can be estimated to be about 20 nm^2 , which is significantly smaller than previous estimates based on cryo-EM structures, but still large compared to the area difference between the open and closed state of the ion channel and, thus, in line with the suggested flattening-induced Piezo activation (Guo and MacKinnon, 2017).

In addition, our elasticity analysis of the nanodome shapes allowed to construct an elastic model for Piezo activation that distinguishes the different energy components in the tension-induced flattening of the protein-membrane nanodome. According to this model, the Piezo protein elasticity can be approximately described as a harmonic spring, with a force-free conformation that agrees with the curved protein conformation in the cryo-EM structures within the modeling accuracy. The elastic energy of the membrane, which opposes the protein elasticity, is the sum of the membrane bending energy E_b and the tension-associated energy term $\gamma \Delta A$ that drives the tension-induced flattening. In tensionless membranes, the interplay of the protein elastic energy E_p with the membrane bending energy E_b leads to a flattening of the protein relative to cryo-EM structure that is associated with a force of roughly 100 pN acting on the protein (see **Figure 5(d)**). With further tension-induced flattening, the force on the protein increases by about 60 pN

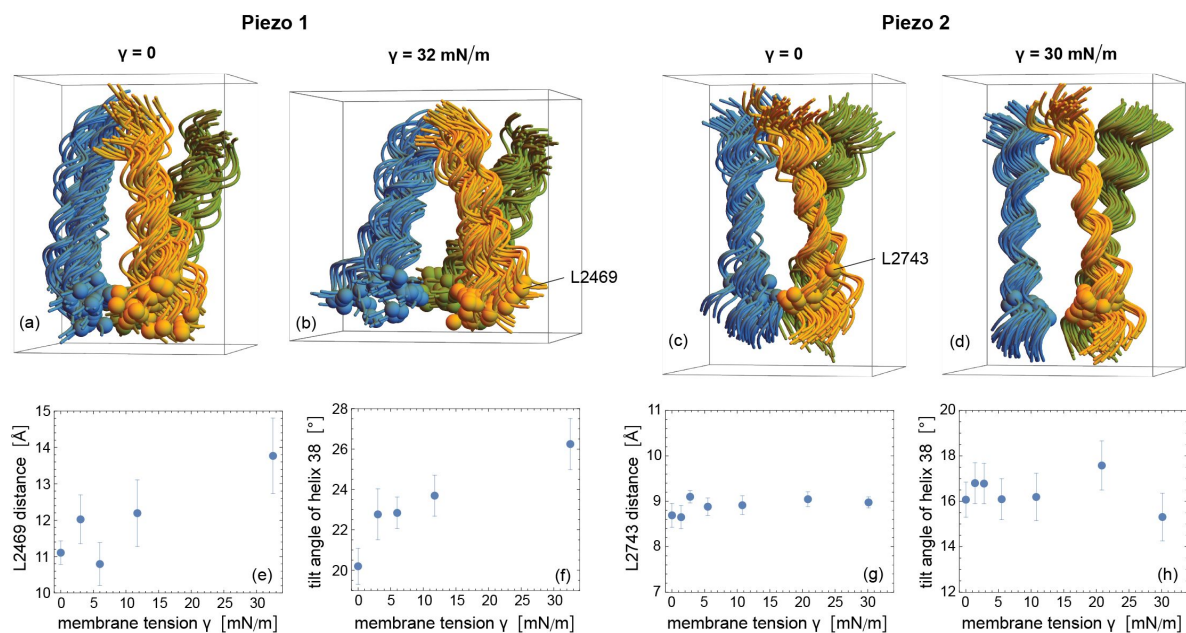


Figure 6.

Response of the Piezo 1 and Piezo 2 channels to membrane tension.

(a,b) Simulation conformations of the three TM helices 38 that line the ion channel from our coarse-grained simulations of membrane-embedded Piezo 1 at vanishing membrane tension $\gamma = 0$ and at the largest simulated tension $\gamma = 32 \text{ nM/m}$. (c,d) Simulation conformations of the TM helices 38 from our coarse-grained simulations of membrane-embedded Piezo 2 at $\gamma = 0$ and the largest simulated tension $\gamma = 30 \text{ nM/m}$. The 50 aligned simulation conformations in (a) to (d) are taken from the last microseconds of the 10 simulation trajectories at intervals of $0.25 \mu\text{s}$ along each trajectory (5 conformations per trajectory), with TM helices 38 depicted as spline curves of backbone atoms in Mathematica 13. (e) Average distance between the backbone beads of the residues L2469 that form the outermost constriction site of the ion channel in Piezo 1 (Yang et al., 2022) versus membrane tension γ of our simulations. The L2496 backbone beads of the three TM helices 38 are shown as beads in (a) and (b). (f) Average tilt angle of the Piezo 1 helices 38 relative to the vertical z direction of the simulation conformations in (a,b) versus membrane tension γ of our simulations. (g) Average distance between the backbone atoms of the residues L2743 that form the outermost constriction site of the ion channel in Piezo 2 (Yang et al., 2022) versus membrane tension γ . (h) Average tilt angle of the Piezo 2 helices 38 versus membrane tension γ . Error bars in (e) to (h) represent errors of the mean calculated from average values obtained for the 10 trajectories.

per nanometer in reduced protein height. With atomic force microscopy (AFM), Lin et al. (2019) [measured](#) the force-induced height change of Piezo1 protein-membrane nanodomains adsorbed to substrates. In this system, the force-induced height change is not only affected by the elastic energy of the protein and the membrane, but also by the adhesion energy of the membrane to the substrate, which led to the spreading of membranes vesicles with embedded Piezo 1 on the substrate in the generation of the substrate-adsorbed membranes, and may also be affected by the water pocket between the nanodome and the substrate. The smaller force response of about 7 pN per nanometer in reduced nanodome height measured in the AFM experiments therefore cannot be compared to the force response of the protein obtained from our modeling of simulation data for the tension-induced flattening of the Piezo protein-membrane nanodome.

Besides the excess area ΔA , our elastic model for Piezo activation is based on the bending energy E_b of the membrane nanodome, which we determined from extrapolations of E_b values for average nanodome shapes in different time intervals of the trajectories (see **Figure 4-figure supplement 1** [Figure 4-figure supplement 1](#)), akin to the extrapolations of ΔA values. Compared to ΔA , the bending energies E_b of non-planar membrane shapes are more sensitive to membrane shape undulations because of the scale invariance of E_b . This scale invariance implies that the bending energy for small perturbations of the planar state is identical to the bending energy of large perturbations with the same shape, and membrane shape undulations can lead to many such small perturbations. However, the membrane shapes obtained from averaging our coarse-grained simulation conformations are rather smooth and do not exhibit undulations (see **Figures 1** [Figure 1](#) and **2** [Figure 2](#)). To corroborate our bending energy calculations for these averaged three-dimensional nanodome shapes, we note that essentially identical bending energies can be obtained from the highly smoothened mean curvatures M of the two-dimensional membrane profiles. For the mean curvature profile M of the relaxed and tensionless Piezo 2 membrane nanodome in **Figure 1(c)** [Figure 1\(c\)](#), we obtain a bending energy of about 1.7κ , which agrees within numerical accuracy with the bending energy value $E_b = 1.93 \pm 0.25 \kappa$ obtained from extrapolation of E_b values for average three-dimensional nanodome shapes (see **Figure 4-figure supplement 1** [Figure 4-figure supplement 1](#)). At membrane tensions $\gamma = 1.4, 2.8, 5.5$, and 10.8 mN/m, we obtain the bending energies of about $1.4, 1.2, 0.8$, and 0.5κ , respectively, from two-dimensional mean-curvature profiles M smoothened as in **Figure 1(c)** [Figure 1\(c\)](#), which all agree within numerical accuracy with the bending energy values in **Figure 5(c)** [Figure 5\(c\)](#) obtained from three-dimensional shapes. In general, bending energies calculated from two-dimensional nanodome profiles can be expected to be slightly lower than the bending energies determined for the three-dimensional average shapes, because these shapes have the three-fold symmetry of the Piezo proteins, but only approximate rotational symmetry (see **Figure 2** [Figure 2](#)).

In tensionless membranes, we obtain a nearly catenoidal shape of the membrane surrounding the Piezo protein nanodome in our coarse-grained simulations, in agreement with calculations for radially symmetric nanodome shapes (Haselwandter and MacKinnon, 2018 [Haselwandter and MacKinnon, 2018](#)). The mean curvature of the non-planar catenoidal shape is zero because of oppositely equal principal curvatures. We obtain a nearly zero mean curvature for radial distances larger than about 17 nm from the channel center of the relaxed Piezo 2 protein structure (see **Figure 1(c)** [Figure 1\(c\)](#)), which indicates that the non-planar membrane profile (see **Figure 1(b)** [Figure 1\(b\)](#)) is catenoidal. For tension values equal to or larger than 2.8 mN/m, in contrast, the membrane profile obtained from our coarse-grained simulations is planar for radial distances larger than about 17 nm from the channel center (see **Figure 2(d)** [Figure 2\(d\)](#)). This planar membrane profile around the Piezo nanodome results from the interplay of the membrane bending energy and the energy $\gamma \Delta A$ associated with the membrane tension γ . A characteristic length scale of this interplay is the crossover length $\sqrt{\kappa/\sigma}$. The membrane tension dominates the elastic energy of the membrane on length scales larger than this crossover length. For a membrane tension $\gamma = 1$ mN/m and a bending rigidity κ of $25 k_B T$, the crossover length is 10 nm. For $\gamma = 4$ mN/m, the crossover length is reduced to 5 nm. A dominance of membrane tension leads to short-range exponential decay of membrane perturbations out of the planar membrane state, in contrast to the long-range logarithmic decay of the catenoid in tensionless membranes. The results of our coarse-grained simulations thus are independent of the

finite membrane size in the simulations are least for tension values equal to or larger than 2.8 mN/m. For $\gamma = 0$, the on average zero membrane slope imposed at the periodic boundaries of the simulation in principle induces deviations from the catenoidal shape. However, these deviations appear to be small for the simulation-box area of about $50 \times 50 \text{ nm}^2$ in our coarse-grained simulations and are not reflected in the mean curvature profile of **Figure 1(c)** with nearly zero mean curvatures from about 17 nm to the largest radial distances of about 25 nm compatible with the simulation box.

The simulation-box area of about $33 \times 33 \text{ nm}^2$ in our atomistic simulations is larger than the simulation box area in previous atomistic simulations (see Introduction), but clearly smaller than the corresponding area of about $50 \times 50 \text{ nm}^2$ in our coarse-grained simulations. In addition, the trajectory lengths up to 300 ns of the computationally significantly more demanding atomistic simulations are clearly shorter than the trajectory lengths between 4 and 8 μs of our coarse-grained simulations. These different membrane areas and trajectory lengths constitute two caveats in the comparison of atomistic and coarse-grained simulation results. To address the smaller size of the atomistic simulations, we first note the excess area ΔA of the tensionless nanodome is dominated by the membrane deformations in the center of the nanodome where the protein is located. In the case of the tensionless average shapes obtained from our coarse-grained simulations, limiting the excess area calculation up to radial distances of 12 nm from the channel center where the outermost 4-TM helix units are located (see **Figure 1(d)**) leads to values that are only about 10% smaller than the values for the entire average shape with maximal radial distance of about 25 nm. Limiting the excess area calculation to radial distances of 16.5 nm, which correspond to the maximal radial distance of the atomistic simulations, leads to values that are only about 3% smaller. The smaller membrane area of our atomistic simulations thus appears to be sufficient to capture the main deformations of the nanodome in tensionless membranes. At membrane tensions of 2 mN/m or larger, the membrane around the Piezo protein is already rather flat and contributes only marginally to the overall excess area ΔA of the nanodome (see profiles in **Figures 2(d)** and **3(d)**). However, for the membrane tensions of 2, 4, and 8 mN/m, the ΔA values obtained from the extrapolation of atomistic simulation results tend to be somewhat larger than the corresponding ΔA values obtained in the coarse-grained simulation systems. These differences in the ΔA values may result from remaining roughnesses in the averaged atomistic conformations (see **Figure 3a** and **b**), from a larger resistance of the atomistic Piezo 2 protein to tension-induced flattening, or from inaccuracies in the extrapolation of ΔA values for the atomistic system. The atomistic simulation trajectories start from a membrane shape obtained in coarse-grained simulations after 200 ns of relaxation (see Methods), which is converted to atomistic resolution. Within these 200 ns of relaxation in coarse-grained simulations, the excess area of 165 nm^2 for the initial nanodome with cryo-EM Piezo 2 structure as starting structure is reduced by about half. In the five equal time windows of the 300 ns atomistic trajectories for $\gamma = 0$, the averaged shapes have the excess areas $\Delta A = 68 \pm 2, 61 \pm 2, 56 \pm 2, 52 \pm 2$, and $51 \pm 2 \text{ nm}^2$. In **Figure 4(c)**, the ΔA values of the last four time windows are plotted as function of the inverse simulation time, i.e. of the inverse center of the time windows. From linear extrapolation of these four data points, we obtain the value $\Delta A = 45 \pm 5 \text{ nm}^2$ for $\gamma = 0$. For $\gamma = 4$ and 8 mN/m, the linear extrapolation curves are steeper because the simulation starts from the same structure as the tensionless simulations, which leads to larger uncertainties and potential overestimates in the extrapolation. For the largest membrane tension $\gamma = 18 \text{ mN/m}$, the excess area is already reduced to about 20 nm^2 in the first two time windows of the simulations, followed by smaller reduction in the remaining three time windows with data points included in **Figure 4(c)**. Overall, the excess area ΔA clearly decreases in all atomistic simulations relative to the initial structure obtained after short relaxation of 200 ns in coarse-grained simulations. The initial relaxation in coarse-grained simulations facilitates the relaxation towards equilibrium in the atomistic simulations.

Materials and methods

Coarse-grained simulations

Protein modeling

Our Piezo 2 simulation system is based on the cryo-EM structure of the full-length Piezo 2 trimer (PDB ID 6KG7) (Wang et al., 2019 [DOI](#)), which includes all 38 TM helices of the Piezo 2 monomers. We added shorter loops of less than 25 residues not resolved in this cryo-EM structure with the software MODELLER v10.2 (Šali and Blundell, 1993 [DOI](#); Webb and Šali, 2016 [DOI](#)). The remaining unresolved longer loops not included in our Piezo 2 simulation system are indicated by dashed lines in **Fig. 7** [DOI](#). In cryo-EM structures of Piezo 1 (Guo and MacKinnon, 2017 [DOI](#); Saotome et al., 2018 [DOI](#); Zhao et al., 2018 [DOI](#)), several of the outer, N-terminal TM helices are not resolved. We therefore used the cryo-EM structure of full-length Piezo2 as template structure to add the unresolved 12 N-terminal TM helices to the Piezo 1 cryo-EM structure with PDB ID 6B3R (Guo and MacKinnon, 2017 [DOI](#)) via homology modelling with MODELLER v10.2. Shorter loops of less than 25 residues not resolved in the resulting structure were added as for Piezo 2. To avoid artefacts from missing long protein loops, the distances between the terminal residues of these missing loops were constrained to the distances in the cryo-EM structures using a harmonic potential with spring constant $10 \text{ kJ mol}^{-1} \text{ nm}^{-2}$ in the coarse-grained simulations.

Force field

We used the Martini 2.2 force field, which exhibits a general 4 to 1 mapping of non-hydrogen atoms into coarse-grained simulation beads (de Jong et al., 2013b [DOI](#); Periole and Marrink, 2013 [DOI](#); Marrink et al., 2007 [DOI](#); Monticelli et al., 2008 [DOI](#)), in all coarse-grained simulations performed with the software GROMACS 2018.3 (Abraham et al., 2015 [DOI](#)). To allow for tension-induced conformational changes of Piezo 2, we employed the standard protein secondary structure constraints of Martini 2.2 with secondary structures identified with DSSP (Kabsch and Sander, 1983 [DOI](#)), but did not constrain the protein tertiary structure by an additional elastic network (Periole and Marrink, 2013 [DOI](#)). Instead, we relied on the capabilities of the Martini coarse-grained force field for modeling membrane systems with TM helix assemblies (Sharma and Juffer, 2013 [DOI](#); Chavent et al., 2014 [DOI](#); Majumder and Straub, 2021 [DOI](#)).

Membrane embedding

To set up the protein-membrane nanodome of the coarse-grained Piezo 2 simulation system, we first embedded an single monomer of the Piezo 2 trimer into a membrane because the transmembrane region of a Piezo monomer is not strongly curved and can be placed reasonably well in a planar membrane. To this end, we converted the Piezo 2 monomer structure into the coarse-grained Martini 2.2 representation (de Jong et al., 2013b [DOI](#); Marrink et al., 2007 [DOI](#); Monticelli et al., 2008 [DOI](#)) using the martinize script (de Jong et al., 2013a [DOI](#)) and generated a coarse-grained planar, asymmetric membrane of area $50 \text{ nm} \times 50 \text{ nm}$ with the composition indicated in **Table 1** [DOI](#) using the INSert membrANE (INSANE) software script (Wassenaar et al., 2015 [DOI](#)). As in previous simulation studies of membrane-embedded Piezo proteins (Buyan et al., 2020 [DOI](#); Chong et al., 2021 [DOI](#); De Vecchis et al., 2021 [DOI](#); Jiang et al., 2021 [DOI](#)), we chose a asymmetric membrane composition that mimics the cell membrane composition with negatively charged lipids in the inner leaflet.

We then oriented and centered the transmembrane domain of the Piezo 2 monomer along the x-y plane of the membrane and solvated the system with coarse-grained water beads in GRO-MACS 2018.3 (Abraham et al., 2015 [DOI](#)). To equilibrate the membrane around the Piezo 2 monomer, we

Figure 7.

Structure and topology of the Piezo 2 monomer (Wang et al., 2019).

The 38 TM helices of the monomer are numbered from the N-to the C-terminus of the protein chain. The TM helices 1 to 36 are arranged in 9 TM-units with 4 helices each. The ion channel of the Piezo 2 trimer is lined by the three TM helices 38 of the three monomers. Loops not resolved in the cryo-EM structure of the Piezo 2 trimer structure in detergent micelles are indicated by dashed lines.

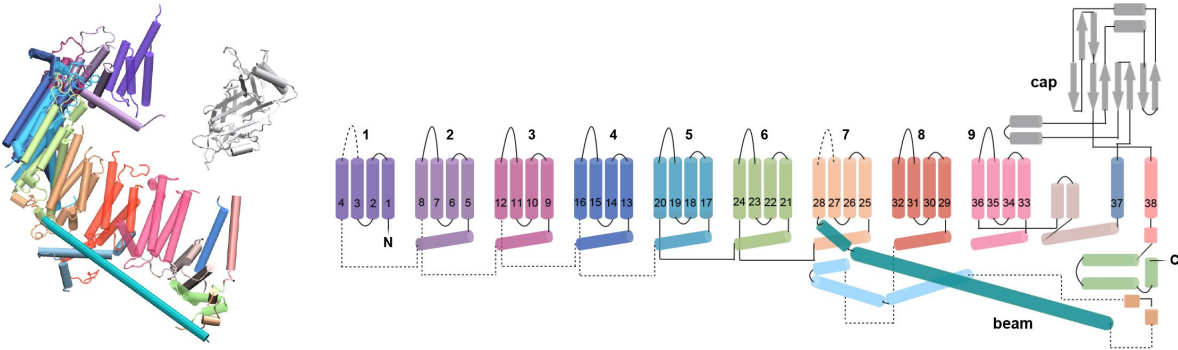


Table 1.

Lipid percentages of membrane

	POPC	POPE	CHOL	POPS	PIP2	DPSM
inner leaflet	50	20	20	5	5	-
outer leaflet	55	20	20	-	-	5

constrained the protein beads using harmonic potentials with force constant $10\,000\text{ kJ mol}^{-1}\text{ nm}^{-2}$, performed an energy minimization with 5000 steps of the steepest descent algorithm, and run a subsequent MD simulation with a length of 50 ns in the NPT ensemble. In this simulation, the pressure was kept at 1 bar using the Berendsen barostat (Berendsen et al., 1984) and the temperature was maintained at 310 K using a velocity-rescale thermostat (Bussi et al., 2007) with individual coupling of protein, membrane, and solvent. The distance cutoff for the non-bonded interactions was 1.2 nm in this simulation, and bonds were constrained with the LINCS algorithm (Hess et al., 1997).

To embed the Piezo 2 trimer, we first generated a Piezo 2 monomer with ‘lipid envelop’ by selecting all lipid molecules of the membrane-embedded protein monomer that contain at least one bead with distance smaller than 3 nm from a protein backbone bead. We then superimposed this lipid-enveloped monomer along each of the three monomers of the Piezo 2 cryo-EM structure with Pymol 2.5.0 (Schrödinger, LLC, 2015) and deleted overlapping lipids from the three envelops. We then placed the resulting lipid-enveloped Piezo 2 trimer into a coarse-grained, planar and asymmetric membrane of area $50\text{ nm} \times 50\text{ nm}$ with the center of mass of the Piezo 2 ion channel about 4 nm below the membrane midplane (see Fig. 8(a)). In this placement, the Piezo 2 trimer remains oriented as in the cryo-EM structure, with the ion channel roughly oriented along the z-axis and, thus, perpendicular to the x-y plane of the membrane. To merge the curved lipid envelop of the Piezo 2 trimer and the planar membrane into a membrane nanodome, we first added water beads and sodium and chloride ions to neutralize the overall system charge at a physiological salt concentration of 150 mM with GROMACS. The system was then energy minimized in 10 000 steps of steepest descent and equilibrated in a 10 ns MD simulation in the NVT ensemble with an integration time step of 10 fs and a subsequent 500 ns MD simulation in the NPT ensemble with an integration time step of 20 fs. In these minimization and equilibrium steps, the protein beads were constrained by a harmonic potential with force constant of $1000\text{ kJ mol}^{-1}\text{ nm}^{-2}$ to maintain the protein shape. The temperature in the equilibration simulations was kept at 310 K by a velocity-rescale thermostat, and the pressure in the NPT simulations was kept at 1 bar with the Berendsen barostat. During these equilibration simulations, a continuous membrane nanodome is formed (see Fig. 8(b)). To embed the Piezo 1 trimer, we took advantage of the structural similarity to Piezo 2 and replaced the membrane-embedded Piezo 2 trimer by the modelled full-length Piezo 1 trimer after structural alignment in Pymol.

Relaxation

After these membrane embedding and lipid equilibration steps, we slowly released the position restraints on the protein atoms by reducing the force constant to 1000, 100, 10 and $1\text{ kJ mol}^{-1}\text{ nm}^{-2}$ in four MD simulation steps of 10 ns each in the NPT ensemble with an integration time step of 20 fs. In these simulations, protein, membrane, and solvent beads were coupled independently to an external bath to keep the temperature at 310 K using a velocity-rescale thermostat (Bussi et al., 2007). The lateral and normal pressure was maintained at 1 bar using semi-isotropic pressure coupling and the Berendsen barostat (Berendsen et al., 1984) with a collision frequency of 5 ps. We implemented bond constraints using the LINCS algorithm (Hess et al., 1997) and employed a cut-off of 1.2 nm for the non-bonded interactions. As last step, we removed the position restraints on protein beads and run 10 independent trajectories of length 200 ns to generate 10 different input structures for the production simulation runs.

Coarse-grained production simulations

In the production simulation runs, we switched to the Parrinello-Rahman barostat (Parrinello and Rahman, 1981) with a coupling constant of 20 ps to achieve more flexibility for box shape modulations with semi-isotropic pressure coupling (Braun et al., 2019). All other simulation parameters were identical to the parameters of the final equilibration simulations. To perform simulations at different membrane tensions, we kept the pressure in the normal z direction at $P_z = 1\text{ bar}$ and applied a series of pressure values in the lateral x and y directions of the membrane

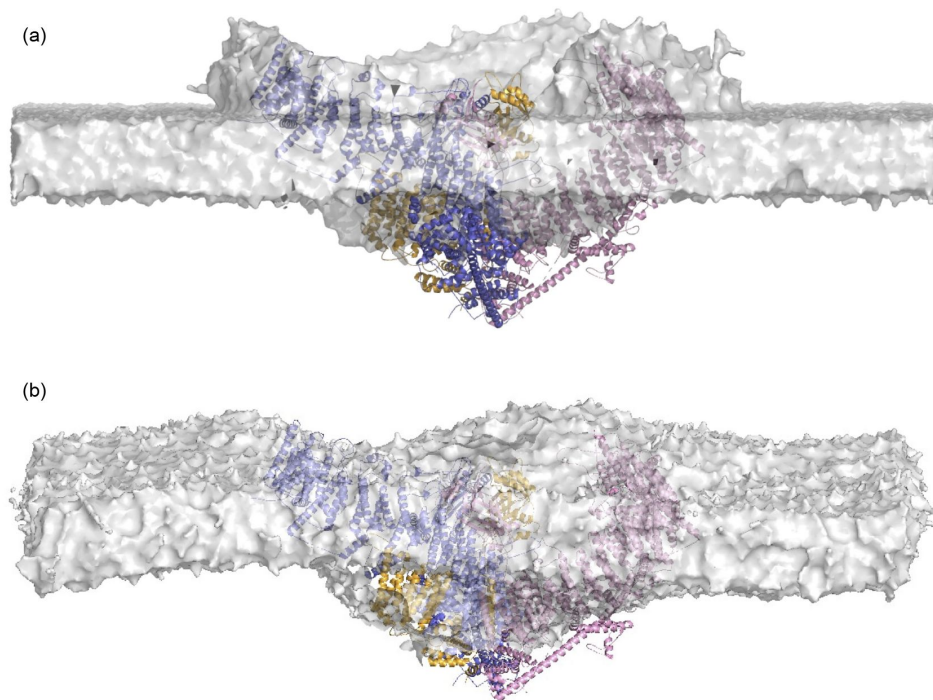


Figure 8.

Embedding the Piezo 2 protein into a membrane.

(a) Piezo 2 trimer with 'lipid envelop' placed into planar and asymmetric membrane of area 50 nm × 50 nm. The lipid envelop was created by insertion of a Piezo 2 monomer into a planar lipid membrane and subsequent superposition of resulting lipid-enveloped monomers on the Piezo 2 cryo-EM structure (see Methods). (b) Piezo 2 trimer embedded in a continuous membrane nanodome after minimization and equilibration simulations starting from (a).

plane. These pressure values were $P_x = P_y = 1, 0.5, 0, -1, -3, -7$, and -11 bar in our coarse-grained Piezo 2 simulations and $P_x = P_y = 1, 0, -1, -3$ and -11 bar in our coarse-grained Piezo 1 simulations. The pressure difference in the lateral and normal directions leads to the membrane tension (Yefimov et al., 2008 [DOI](#))

$$\gamma = L_z (P_z - (P_x + P_y)/2) \quad (3)$$

where L_z is the average box height during the simulations. At each lateral pressure, we generated 10 independent production trajectories starting from the same initial conformation obtained after relaxation with a length of 4 μ s each. For Piezo 2, we extended the 10 simulation trajectories at zero membrane tension to 8 μ s to explore the role of the long-range shape fluctuations in tensionless membranes in the equilibration.

Atomistic simulations

System setup

To set up our atomistic simulations of membrane-embedded Piezo 2, we used a coarse-grained Piezo 2 conformation obtained after 200 ns of final equilibration as starting point. The coarse-grained Piezo 2 trimer and surrounding membrane nanodome of this conformation were converted to atomistic resolution with the CHARMM-GUI server (Wassenaar et al., 2014 [DOI](#); Jo et al., 2008 [DOI](#); Brooks et al., 2009 [DOI](#); Lee et al., 2016 [DOI](#)). In this conversion, the phosphoinositol biphosphate (PIP2) of the coarse-grained Martini model was mapped to phosphoinositol (4,5) biphosphate in the atomistic model. We reduced the membrane area to about $33 \times 33 \text{ nm}^2$ to decrease the total number of atoms in the system and removed atomic clashes of lipids and amino acids by energy minimization with 5000 steps of the steepest descent algorithm in GROMACS. We then solvated and neutralized the system at a total salt concentration of 150 mM of KCl by adding an appropriate number of potassium and chloride ion. In our atomistic simulations, we used the CHARMM36 force field (Huang and MacKerell Jr, 2013 [DOI](#); Klauda et al., 2010 [DOI](#)) with the TIP3P water model (Jorgensen et al., 1983 [DOI](#)).

System equilibration

We performed the equilibration and production simulations with the Amber20 software (Case et al., 2020 [DOI](#)) because of the efficient use of GPUs with this software (Salomon-Ferrer et al., 2013 [DOI](#)). To this end, we first generated Amber suitable input coordinates using the ParmEd software tool (Shirts et al., 2017 [DOI](#)) and energy-minimized the system with 10 000 steps of the steepest descent algorithm and subsequent 10 000 steps of the conjugate gradient algorithm. We then heated the system from temperature 0 K to 310 K in three simulation steps of 10 ns in the NVT ensemble using an integration time step of 1 fs and a Langevin thermostat with a collision frequency of 1 ps^{-1} . In these three heating simulations, the positions of the protein backbone atoms were restrained by harmonic potentials with a force constant of $10 \text{ kcal mol}^{-1} \text{ \AA}^{-2}$. We subsequently released the harmonic restraints in 5 simulation steps of 10 ns in the NPT ensemble, decreasing the force constant by a factor of 2 in each of the steps. In these equilibration simulations with an integration time step of 2 fs, the temperature was maintained at 310 K by a Langevin thermostat with a collision frequency of 1 ps^{-1} , the pressure in normal and lateral directions was kept at 1 bar by a Berendsen barostat with a semi-isotropic pressure coupling and a pressure relaxation time of 2 ps, a cutoff length of 10 $\text{\AA}$ was used for non-bonded interactions, long-range electrostatic interactions were calculated with the Particle Mesh Ewald (PME) method (Darden et al., 1993 [DOI](#); Es-smann et al., 1995 [DOI](#)), and all bonds involving hydrogen atoms were constrained using the SHAKE algorithm (Ryckaert et al., 1977 [DOI](#)).

Production simulations

In the production simulations starting from the equilibrated system conformation, we used hydrogen mass repartitioning (Hopkins et al., 2015 [DOI](#)) to increase the integration timestep to 4 fs for computational efficiency. Our production simulations consist of 5 independent trajectories at each of the membrane tension values 0, 1, 2, 4, 8, and 18 mN/M. In our simulations with the Amber20 software, the membrane tension was implemented by subjecting a constant surface tension at the two interfaces of the lipid bilayer in the *x-y* plane, using the Berendsen barostat with a reduced pressure relaxation time of 1 ps to ensure rapid pressure regulation under constant surface tension conditions. As in the equilibration simulations, the temperature was maintained at 310K by a Langevin thermostat with a collision frequency of 1 ps^{-1} , bonds containing hydrogen atoms were constrained using the SHAKE algorithm, a cutoff length of 10 Å was used for non-bonded interactions, and long-range electrostatic interactions were calculated with the PME method. At each of tension values 0, 2, 4, and 8 mN/M, we generated 5 independent trajectories with a trajectory length of 300 ns starting from the same initial structure. At the tension values 1 and 18 mN/M, we generated 5 trajectories with a length of 180 ns each.

Analysis of trajectories

Membrane shape of simulation conformations

To determine the membrane midplane shape of a coarse-grained simulation conformation, we first translated the conformation so that the center of mass of the ion channel formed by the three helices 38 is located at the center of the simulation box. We next discretized the *x-y* plane of the simulation box into a square lattice with 25×25 equally sized squares (bins), separated the lipid head beads (head beads of POPC, POPE, POPS, POP2 and DPSM) into upper and lower membrane leaflet, and allocated these head beads to bins based on the *x-y* coordinates of the beads. In each bin with at least two lipid head beads in the upper and in the lower leaflet, we calculated the average *z* coordinate z_{upper} of the lipid head beads in the upper leaflet in this bin, the average *z* coordinate z_{lower} of the lipid head beads in the lower leaflet, and the midplane coordinate z_{midplane} of the bin as the average of z_{upper} and z_{lower} . We finally performed a bilinear interpolation of the calculated midplane coordinates of the square lattice with Mathematica 13 to determine *z* coordinates also for bins with less than two lipid head beads in the upper or lower leaflet, and to obtain continuous midplane shapes. We determined these membrane midplane shapes for simulations frames at intervals of 20 ns, so for e.g. 201 frames of a coarse-grained simulation trajectory with a length of 4 μs. The membrane midplane shape of atomistic simulation conformations was calculated similarly, for a square lattice of 17×17 equally sized bins because of the smaller membrane size in our atomistic simulations, using the coordinates of the common phosphorus atoms in the lipid heads for bin allocation and averaging. Along the atomistic simulation trajectories, we determined membrane midplane shapes for simulation frames at intervals of 1 ns.

Average membrane shapes and lipid densities

To calculate average membrane midplane shapes from the continuous midplane shapes of individual conformations, we first aligned the membrane shapes by rotations around an axis in *z*-direction through the center of mass of the ion channel in the center of the simulation box. In this rotational alignment, the rotation angles were chosen so that the *x-y* coordinates of the centres of mass of the three helices 38 forming the channel in the underlying simulation conformations were aligned. To generate average shapes that reflect the 3-fold symmetry of the Piezo proteins, we performed additional rotations by 120 and 240 degrees of all rotationally aligned membrane shapes and included the two additional resulting shapes per membrane shape in the averaging. In our figures and calculations, the average membrane shapes have a circular projected area with a

diameter d_m that corresponds to the average x and y dimensions of the simulation boxes of the conformations, because only these ‘inscribed’ circular membrane segments are overlying in all rotated conformations. In an analogous way, we calculated average, 3-fold symmetric lipid densities from the interpolated lipid numbers in the bins of the square lattice of each conformation.

Excess area and bending energy

To numerically determine the excess area and bending energy of average membrane midplane shapes, we discretized the continuous average shapes and lipid densities obtained after the interpolation, rotation, and averaging steps described above by discretizing the reference x - y plane of the Monge parametrization into a square lattice with lattice constant $a = 1$ nm. This lattice constant is chosen to be smaller than the bin width of about 2 nm used in determining the membrane shape of the simulation conformations, to take into account that the averaging of these membrane shapes can lead to a higher resolution compared to the 2 nm resolution of the individual membrane shapes. The excess area at site (x, y) was calculated as

$$\delta A(x, y) = a^2 \left(\sqrt{1 + (z(x+a, y) - z(x, y))^2/a^2 + (z(x, y+a) - z(x, y))^2/a^2} - 1 \right) \quad (4)$$

where $z(x, y)$ denotes the midplane z coordinate at site (x, y) , and the overall excess area ΔA of the average shape was obtained by summing up the excess areas $\delta A(x, y)$ at all sites of the shape, i.e. at all sites with $x^2 + y^2 \leq (d_m/2)^2$. To compensate for incomplete equilibration of the membrane shapes within the simulation times of the trajectories, we divide the trajectories into 5 equal time intervals, determine the excess area ΔA of the nanodome shape obtained from averaging over conformations of all independent simulation runs in this time interval, and linearly extrapolate the ΔA values to long timescales as shown in **Figure 4**. The ΔA values obtained from these extrapolations are plotted in **Figure 3(e)**. For a lattice constant of $a = 2$ nm, we obtain extrapolated values of the excess area ΔA from the coarse-grained simulations that are 2 to 3% lower than the values for $a = 1$ nm, which is a small compared to statistical uncertainties with relative errors of around 10%. From the atomistic simulations of the Piezo 2 protein-membrane nanodome, we obtain extrapolated values of ΔA for a lattice constant of $a = 2$ nm that are 3 to 6% lower than the values for $a = 1$ nm.

We determined the bending energy at site (x, y) based on the general expression for the mean curvature H in Monge parametrization as

$$e_b(x, y) = 2\kappa H(x, y)^2 dA(x, y) = \kappa a^2 \frac{z_{xx}(1 + z_y^2) + z_{yy}(1 + z_x^2) - 2z_{xy}z_xz_y}{2(1 + z_x^2 + z_y^2)^{5/2}} \quad (5)$$

with bending rigidity κ and the standard discretizations of the partial derivatives

$$z_x = (z(x+a, y) + z(x-a, y))/2a, \quad z_y = (z(x, y+a) + z(x, y-a))/2a \quad (6)$$

$$z_{xx} = (z(x+a, y) + z(x-a, y) - 2z(x, y))/a^2, \quad z_{yy} = (z(x, y+a) + z(x, y-a) - 2z(x, y))/a^2 \quad (7)$$

$$z_{xy} = (z(x+a, y+a) + z(x-a, y-a) - z(x+a, y-a) - z(x-a, y+a))/4a^2 \quad (8)$$

The overall bending energy of the average membrane midplane shape of the membrane in the protein-membrane nanodome was then calculated as

$$E_b = \sum_{x,y} e_b(x, y) \rho(x, y) \quad (9)$$

where $\rho(x, y) \leq 1$ is the relative average lipid density at site (x, y) , calculated as the number of lipid head beads at site (x, y) divided by the mean number of lipid head beads for ‘pure’ membrane patches far away from the protein. To compensate for incomplete equilibration, the bending energy E_b of the average membrane shapes obtained in the different time intervals of our coarse-grained simulations are extrapolated to long timescales akin to the excess area ΔA (see **Figure 4 — figure supplement 1** [↗](#)). For a lattice constant of $a = 2$ nm, we obtain extrapolated values of E_b from the coarse-grained simulations of Piezo 2 at the tension values $\gamma = 0, 1.4, 2.8, 5.5$, and 10.8 mN/m that are about 10 to 13% lower than the values for $a = 1$ nm. These lower values of E_b for $a = 2$ nm likely result from the lower resolution of the membrane curvature in the averaged nanodome, rather than from remaining membrane roughness, because the averaged nanodome shapes obtained from our coarse-grained simulations are rather smooth (see **Figures 1a** [↗](#) and **2a,b** [↗](#)). The averaged nanodome shapes obtained from our atomistic simulations, in contrast, exhibit remaining roughness (see **Figure 3a,b** [↗](#)). In contrast to the excess area ΔA , the bending energy is scale invariant and, thus, rather sensitive to remaining roughness on small length scales. We therefore do not determine bending energies for the nanodome shapes obtained from our atomistic simulations.

TM helices

In determining the average COMs of the 4-TM units in **Figures 1(d)** [↗](#) and **2(e)** [↗](#), we discarded residues of TM helix ends that protrude out of the membrane. To identify these residues for the Piezo 1 and Piezo 2 TM helices, we calculated the average contact probability of all TM helix residues with lipid tails in the final two microseconds of the 10 coarse-grained trajectories with tensionless membranes. A TM helix residue was taken to be in contact with lipids tails in a simulation frame if the smallest distance between the beads of this residue to lipid tail beads was less than 0.6 nm. We discarded consecutive stretches of residues at the helix ends with an average contact probability of less than 10% as helix ends that protrude out of the membrane.

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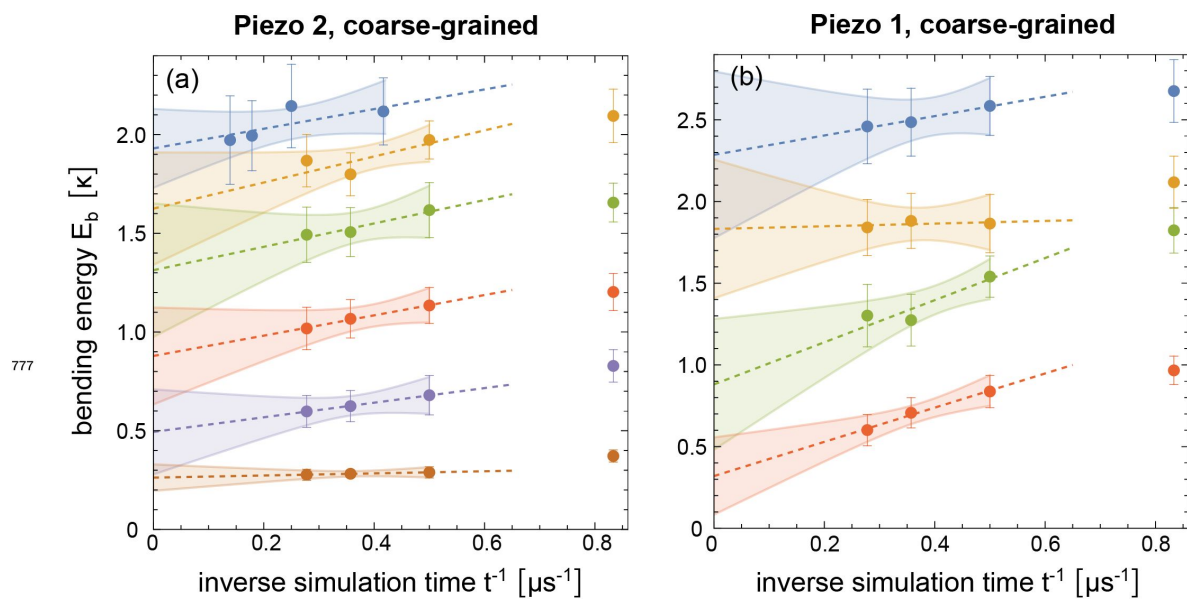


Figure 4—figure supplement 1.

Extrapolation of the membrane bending energy E_b of the nanodome in coarse-grained simulations with (a) Piezo 2 and (b) Piezo 1 to long timescales, akin to the extrapolations of ΔA in Figure 4.

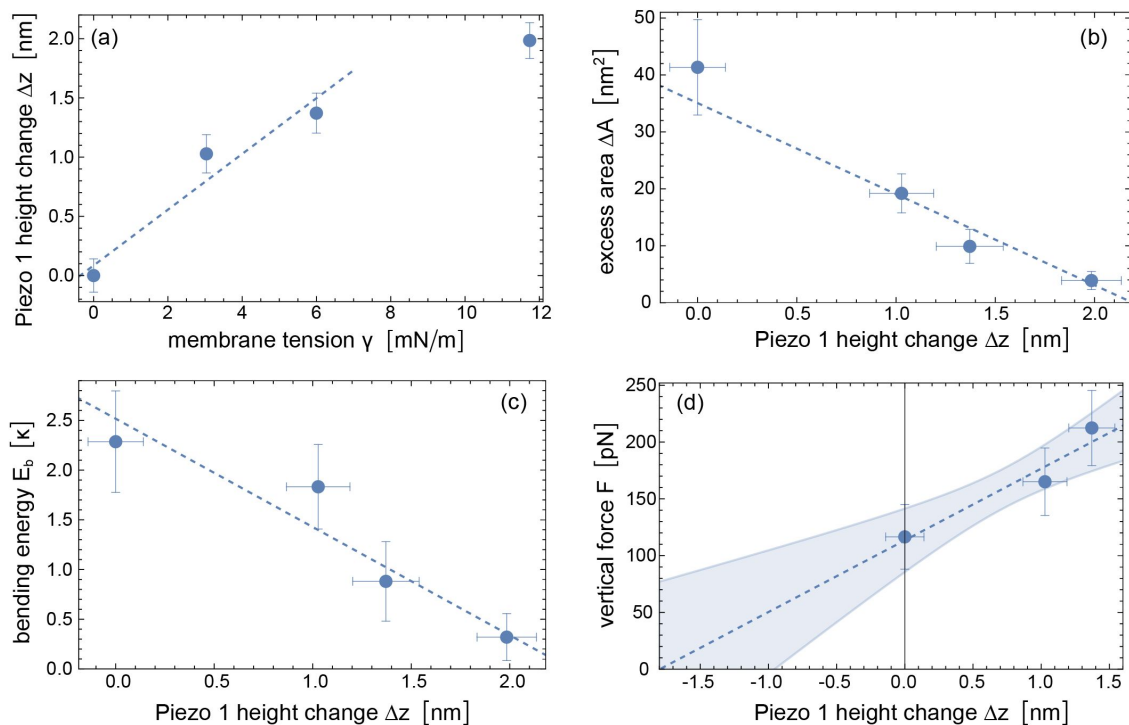


Figure 5—figure supplement 1.

Elasticity modelling of the Piezo 1 protein based on coarse-grained simulation data.

(a) Piezo 1 height change Δz versus membrane tension γ in our coarse-grained simulations of membrane-embedded Piezo 1. The protein height change is defined as $\Delta z(\gamma) = z_4(0) - z_4(\gamma)$ where z_4 is the vertical displacement of the fourth 4-TM unit of the Piezo arms relative to the channel center. (b) Excess area ΔA of the nanodome and (c) bending energy E_b of the lipid membrane in the nanodome versus Piezo height change Δz . The values and errors of ΔA and E_b are obtained from extrapolating simulation results at the tension values $\gamma = 0, 3.0, 6.0$, and 11.8 to long timescales (see **Figure 4(b)** and **Figure 4—figure supplement 1 (b)**). (d) Vertical force F determined from Equation (2) and the slopes of the fitted lines in (b) and (c) at the Δz values obtained from simulations at the tension values $\gamma = 0, 3.0$, and 6.0 . The dashed lines result from linear fitting of all data points in the plots with the function LinearModelFit of Mathematica 13. The shaded error region of the linear fit in (d) represents prediction bands with confidence level 0.5.

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Reviewer #1 (Public review):

Dixit, Noe, and Weikl apply coarse-grained and all-atom molecular dynamics to determine the response of the mechanosensitive proteins Piezo 1 and Piezo 2 proteins to tension. Cryo-EM structures in micelles show a high curvature of the protein whereas structures in lipid bilayers show lower curvature. Is the zero-stress state of the protein closer to the micelle structure or the bilayer structure? Moreover, while the tension sensitivity of channel function can be inferred from experiment, molecular details are not clearly available. How much does the protein's height and effective area change in response to tension? With these

in hand, a quantitative model of its function follows that can be related to the properties of the membrane and the effect of external forces.

Simulations indicate that in a bilayer the protein relaxes from the highly curved cryo-EM dome (Figure 1).

Under applied tension the dome flattens (Figure 2) including the underlying lipid bilayer. The shape of the system is a combination of the membrane mechanical and protein conformational energies (Eq. 1). The membrane mechanical energy is well-characterized. It requires only the curvature and bending modulus as inputs. They determine membrane curvature and the local area metric (Eq. 4) by averaging the height on a grid and computing second derivatives (Eqs. 7, 8) consistent with known differential geometric formulas.

While I am still critical generally of a precise estimate of the energy from simulated membrane shapes (after all it is not trivial to precisely determine even the bending modulus from a simulation), I believe with their revision the authors have convinced me that their estimate is a high quality one, without obvious issues. Although there appears to have been a miscommunication about increasing the density of grain or lowering the density of grain, the authors have tried two grains and determined a similar deformation energy, which addresses my concern. Furthermore, they have computed a dramatically reduced simplification of the curve and determined a similar value.

In summary, this paper uses molecular dynamics simulations to quantify the force of the Piezo 1 and Piezo 2 proteins on a lipid bilayer using simulations under controlled tension, observing the membrane deformation, and using that data to infer protein mechanics. While much of the physical mechanism was previously known, the study itself is a valuable quantification.

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Reviewer #2 (Public review):

Summary:

In this study the authors suggest that the structure of Piezo2 in a tensionless simulation is flatter compared to the electron microscopy structure. This is an interesting observation and highlights the fact that the membrane environment is important for Piezo2 curvature. Additionally, the authors calculate the excess area of Piezo2 and Piezo1, suggesting that it is significantly smaller compared the area calculated using the EM structure or simulations with restrained Piezo2. Finally, the authors propose an elastic model for Piezo proteins. Those are very important findings, which would be of interest to the mechanobiology field.

Whilst I like the suggestion that the membrane environment will change Piezo2 flatness, could this be happening because of the lower resolution of the MARTINI simulations? In other words, would it be possible that MARTINI is not able to model such curvature due to its lower resolution?

Related to my comment above, the authors say that they only restrained the secondary structure using an elastic network model. Whilst I understand why they did this, Piezo proteins are relatively large. How can the authors know that this type of elastic network model restrains, combined with the fact that MARTINI simulations are perhaps not very accurate in predicting protein conformations, can accurately represent the changes that happen within Piezo channel during membrane tension?

Modelling or Piezo1, seems to be based on homology to Piezo2. However, the authors need to further evaluate their model, e.g. how it compares with an AlphaFold model.

To calculate the tension induce flattening of Piezo channel, the authors "divide all simulation trajectories into 5 equal intervals and determine the nanodome shape in each interval by averaging over the conformations of all independent simulation runs in this interval.". However, probably the change in the flattening of Piezo channel happens very quickly during the simulations, possibly within the same interval. Is this the case? and if yes does this affects their calculations?

Finally, the authors use a specific lipid composition, which is asymmetric. Is it possible that the asymmetry of the membrane causes some of the changes in the curvature that they observe? Perhaps more controls, e.g. with a symmetric POPC bilayer is needed to identify whether membrane asymmetry plays a role in the membrane curvature they observe.

<https://doi.org/10.7554/eLife.105138.2.sa2>

Reviewer #3 (Public review):

Strengths:

This work focuses on a problem of deep significance: quantifying the structure-tension relationship and underlying mechanism for the mechanosensitive Piezo 1 and 2 channels. Such an objective is challenging for molecular dynamics simulations, due to the relatively large size of each membrane-protein system. Nonetheless, the approach chosen here is based on methodology that is, in principle, established and widely accessible. Therefore, another group of practitioners would likely be able to reproduce these findings with reasonable effort.

More specifically, while acknowledging the limitations of the MARTINI force field, this work makes a significant improvement compared to previous simulations of Piezo proteins by adopting a range of membrane tensions that includes physiologically relevant values (below 10 mN/m).

Weaknesses:

The two main results of this paper are (1) that both channels exhibit a flatter structure compared to cryo-EM measurements, and (2) their estimated force vs. displacement relationship. Although the former correlates at least quantitatively with prior experimental work, the latter relies exclusively on simulation results and model parameters.

My remaining technical concerns in the revised manuscript are as follows:

(1) At each membrane tension, all concurrent atomistic simulations were initialized from the same snapshot of a previous CG simulation: in my opinion, it is inaccurate to refer to those atomistic simulations as "independent" from each other (as is done twice in the caption of Figure 3, as well as in the text).

(2) Continuum mechanics calculations were employed to model the membrane's curvature energetics. The bending modulus, k , was not determined for the specific lipid composition used in this study, but was instead taken from previous MARTINI simulations involving the same primary lipid, POPC. Given that these calculations are intended to describe MARTINI simulations specifically, this approximation may be acceptable. However, it does not account for the increased stiffness observed in POPC/cholesterol mixtures—an effect measured experimentally but not reproduced by the MARTINI model—nor does it reflect the asymmetric conditions, as all referenced simulations involve symmetric bilayers. As a result, the bending energies and forces shown in Figure 5(c,d) are internally consistent within the model, but they probably correspond to real values up to an unknown multiplicative factor.

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Dixit, Noe, and Weikl apply coarse-grained and all-atom molecular dynamics to determine the response of the mechanosensitive proteins Piezo 1 and Piezo 2 proteins to tension. Cryo-EM structures in micelles show a high curvature of the protein whereas structures in lipid bilayers show lower curvature. Is the zero-stress state of the protein closer to the micelle structure or the bilayer structure? Moreover, while the tension sensitivity of channel function can be inferred from the experiment, molecular details are not clearly available. How much does the protein's height and effective area change in response to tension? With these in hand, a quantitative model of its function follows that can be related to the properties of the membrane and the effect of external forces.

Simulations indicate that in a bilayer the protein relaxes from the highly curved cryo-EM dome (Figure 1).

Under applied tension, the dome flattens (Figure 2) including the underlying lipid bilayer. The shape of the system is a combination of the membrane mechanical and protein conformational energies (Equation 1). The membrane's mechanical energy is well-characterized. It requires only the curvature and bending modulus as inputs. They determine membrane curvature and the local area metric (Equation 4) by averaging the height on a grid and computing second derivatives (Equations 7, 8) consistent with known differential geometric formulas.

The bending energy can be limited to the nano dome but this implies that the noise in the membrane energy is significant. Where there is noise outside the dome there is noise inside the dome. At the least, they could characterize the noisy energy due to inadequate averaging of membrane shape.

My concern for this paper is that they are significantly overestimating the membrane deformation energy based on their numerical scheme, which in turn leads to a much stiffer model of the protein itself.

We agree that “thermal noise” is intrinsic to MD simulations, as in “real” systems, leading to thermally excited shape fluctuations of membranes and conformational fluctuations of proteins. However, for our coarse-grained simulations, the thermally excited membrane shape fluctuations can be averaged out quite well, and the resulting average shapes are smooth, see e.g. the shapes and lines of the contour plots in Fig. 1 and 2. For our atomistic simulations, the averaged shapes are not as smooth, see Fig. 3a and the lines of the contour plots in Fig. 3b. Therefore, we do not report bending energies for the nanodome shapes determined from atomistic simulations, because bending energy calculations are sensitive to remaining “noise” on small scales (due to the scale invariance of the bending energy), in contrast to calculations of excess areas, which we state now on lines 620ff.

For our coarse-grained simulations, we now corroborate our bending energy calculations based on averaged 3d shapes by comparing to bending energy values obtained from highly smoothened 2d mean curvature profiles (see Fig. 1c for mean curvature profiles in tensionless membranes). We discuss this in detail from line 323 on, starting with:

“To corroborate our bending energy calculations for these averaged three-dimensional nanodome shapes, we note that essentially identical bending energies can be obtained from the highly smoothed mean curvatures M of the two-dimensional membrane profiles. ...”

Two things would address this:

(1) Report the membrane energy under different graining schemes (e.g., report schemes up to double the discretization grain).

There are two graining schemes in the modeling, and we have followed the reviewer’s recommendation regarding the second scheme. In the first, more central graining scheme, we use quadratic membrane patches with a sidelength of about 2 nm to determine membrane midplane shapes and lipid densities of each simulation conformation. This graining scheme has also been previously employed in Hu, Lipowsky, Weikl, PNAS 38, 15283 (2013) to determine the shape and thermal roughness of coarse-grained membranes. A sidelength of 2 nm is necessary to have sufficiently many lipid headgroups in the upper and lower leaflet in the membrane patches for estimating the local height of these leaflets, and the local membrane midplane height as average of these leaflet heights (see subsection “Membrane shape of simulation conformation” in the Methods section for details). However, we strongly believe that doubling the sidelength of membrane patches in this discretization is not an option, because a discretization length of 4 nm is too coarse to resolve the membrane deformations in the nanodome, see e.g. the profiles in Fig. 1b. Moreover, any “noise” from this discretization is rather completely smoothed out in the averaging process used in the analysis of the membrane shapes, at least for the coarse-grained simulations. This averaging process requires rotations of membrane conformations to align the protein orientations of the conformations (see subsection “Average membrane shapes and lipid densities” for details). Because of these rotations, the original discretization is “lost” in the averaging, and a continuous membrane shape is generated. To calculate the excess areas and bending energies for this smooth, continuous membrane shape, we use a discretization of the Monge plane into a square lattice with lattice parameter 1 nm. As a response to the referee’s suggestion, we now report that the results for the excess area do not change significantly when doubling this lattice parameter to 2 nm. On line 597, we write:

“For a lattice constant of $a=2$ nm, we obtain extrapolated values of the excess area ΔA from the coarse-grained simulations that are 2 to 3% lower than the values for $a=1$ nm, which is a small compared to statistical uncertainties with relative errors of around 10%.”

On lines 614ff, we now state that the bending energy results are about 10% to 13% lower for $a=2$ nm, likely because of the lower resolution of the curvature in the nanodome compared to $a=1$ nm, rather than incomplete averaging and remaining roughness of the coarse-grained nanodome shapes.

(2) For a Gaussian bump with $\sigma=6$ nm I obtained a bending energy of 0.6κ , so certainly in the ballpark with what they are reporting but significantly lower (compared to 2κ , Figure 5 lower left). It would be simpler to use the Gaussian approximation to their curves in Figure 3 - and I would argue more accurate, especially since they have not reported the variation of the membrane energy with respect to the discretization size and so I cannot judge the dependence of the energy on discretization. I view reporting the variation of the membrane energy with respect to discretization as being essential for the analysis if their goal is to provide a quantitative estimate for the force of Piezo. The Helfrich energy computed from an analytical model with a membrane shape closely resembling the simulated shapes would be very helpful. According to my intuition, finite-difference estimates of curvatures will tend to be overestimates of the true membrane

deformation energy because white noise tends to lead to high curvature at short-length scales, which is strongly penalized by the bending energy.

Instead of Gaussian bumps, we now calculate the membrane bending energy also from the two-dimensional, continuous mean curvature profiles (see Fig. 1c). These mean curvature profiles are highly smoothened (see figure caption for details). Nonetheless, we obtain essentially the same bending energies as in our discrete calculations of averaged, smoothened threedimensional membrane shapes, see new text on lines 326ff. We believe that this agreement corroborates our bending energy calculations. We still focus on values obtained for threedimensional membrane shapes, because of incomplete rotational symmetry. The three-dimensional membrane shapes exhibit variations with the three-fold symmetry of the Piezo proteins, see Figure 2a and b.

We agree that the bending energy of thermally rough membranes depends on the discretization scheme, because the discretization length of any discretization scheme leads to a cut-off length for fluctuation modes in a Fourier analysis. But again, we average out the thermal noise, for reasons given in the Results section, and analyse smooth membrane shapes.

The fitting of the system deformation to the inverse time appears to be incredibly ad hoc ... Nor is it clear that the quantified model will be substantially changed without extrapolation. The authors should either justify the extrapolation more clearly (sorry if I missed it!) or also report the unextrapolated numbers alongside the extrapolated ones.

We report the values of the excess area and bending energy in the different time intervals of our analysis as data points in Fig. 4 with supplement. We find it important to report the time dependence of these quantities, because the intended equilibration of the membrane shapes in our simulations is not “complete” within a certain time window of the simulations. So, just “cutting” the first 20 and 50% of the simulation trajectories, and analysing the remaining parts as “equilibrated” does not seem to be a reasonable choice here, at least for the membrane properties, i.e. for the excess area and bending energy. We agree that the linear extrapolation used in our analysis is a matter of choice. At least for the coarse-grained simulations, the extrapolated values of excess areas and bending energies are rather close to the values obtained in the last time windows (see Figure 4).

In summary, this paper uses molecular dynamics simulations to quantify the force of the Piezo 1 and Piezo 2 proteins on a lipid bilayer using simulations under controlled tension, observing the membrane deformation, and using that data to infer protein mechanics. While much of the physical mechanism was previously known, the study itself is a valuable quantification. I identified one issue in the membrane deformation energy analysis that has large quantitative repercussions for the extracted model.

Reviewer #2 (Public review):

Summary:

In this study, the authors suggest that the structure of Piezo2 in a tensionless simulation is flatter compared to the electron microscopy structure. This is an interesting observation and highlights the fact that the membrane environment is important for Piezo2 curvature. Additionally, the authors calculate the excess area of Piezo2 and Piezo1, suggesting that it is significantly smaller compared to the area calculated using the EM structure or simulations with restrained Piezo2. Finally, the authors propose an elastic model for Piezo proteins. Those are very important findings, which would be of interest to the mechanobiology field.

Whilst I like the suggestion that the membrane environment will change Piezo2 flatness, could this be happening because of the lower resolution of the MARTINI simulations? In other words, would it be possible that MARTINI is not able to model such curvature due to its lower resolution?

Related to my comment above, the authors say that they only restrained the secondary structure using an elastic network model. Whilst I understand why they did this, Piezo proteins are relatively large. How can the authors know that this type of elastic network model restrains, combined with the fact that MARTINI simulations are perhaps not very accurate in predicting protein conformations, can accurately represent the changes that happen within the Piezo channel during membrane tension?

These questions regarding the reliability of the Martini model are very reasonable and are the reason why we include also results from atomistic simulations, at least for Piezo 2, and compare the results. In the Martini model, secondary structure constraints are standard. In addition, constraints on the tertiary structure (e.g. via an elastic network model) are also typically used in simulations of soluble, globular proteins. However, such tertiary constraints would make it impossible to simulate the tension-induced flattening of the Piezo proteins. So instead, as we write on lines 427ff, “we relied on the capabilities of the Martini coarse-grained force field for modeling membrane systems with TM helix assemblies (Sharma and Juffer, 2013; Chavent et al., 2014; Majumder and Straub, 2021).” In these references, Martini simulations were used to study the assembly of transmembrane helices, leading to agreement with experimentally observed structures. As we state in our article, our atomistic simulations corroborate the Martini simulations, with the caveats that are now more extensively discussed in the new last paragraph of the Discussion section starting on line 362.

Modelling of Piezo1, seems to be based on homology to Piezo2. However, the authors need to further evaluate their model, e.g. how it compares with an AlphaFold model.

We understand the question, but see it beyond the scope of our article, also because of the computational demand of the simulations. The question is: Do coarse-grained simulations of Piezo1 based on an AlphaFold model as starting structure lead to different results? It is important to note that we only model the rather flexible 12 TM helices at the outer ends of the Piezo 1 monomers via homology modeling to the Piezo 2 structure, which includes these TM helices. For the inner 26 TM helices, including the channel, we use the high-quality cryo-EM structure of Piezo 1. AlphaFold may be an alternative for modeling the outer 12 helices, but we don't think this would lead to statistically significant differences in simulations – e.g. because of the observed overall agreement of membrane shapes in all our Piezo 1 and Piezo 2 simulation systems.

To calculate the tension-induced flattening of the Piezo channel, the authors “divide all simulation trajectories into 5 equal intervals and determine the nanodome shape in each interval by averaging over the conformations of all independent simulation runs in this interval.”. However, probably the change in the flattening of Piezo channel happens very quickly during the simulations, possibly within the same interval. Is this the case? and if yes does this affect their calculations?

Unfortunately, the flattening is not sufficiently quick, so is not complete within the first time windows, see data points in Figure 4. We therefore report the time dependence with the plots in Figure 4 and extrapolate, see also our response above to reviewer 1.

Finally, the authors use a specific lipid composition, which is asymmetric. Is it possible that the asymmetry of the membrane causes some of the changes in the curvature that they observe? Perhaps more controls, e.g. with a symmetric POPC bilayer are needed to

identify whether membrane asymmetry plays a role in the membrane curvature they observe.

Because of the rather high computational demands, such controls are beyond our scope. We don't expect statistically significant differences for symmetric POPC/cholesterol bilayers. On lines 229ff, we now state:

“Our modelling assumes that any spontaneous curvature from asymmetries in the lipid composition is small compared to the curvature of the nanodome and, thus, negligible, which is plausible for the rather slight lipid asymmetry of our simulated membranes (see Methods).”

Reviewer #3 (Public review):

Strengths:

This work focuses on a problem of deep significance: quantifying the structure-tension relationship and underlying mechanism for the mechanosensitive Piezo 1 and 2 channels. This objective presents a few technical challenges for molecular dynamics simulations, due to the relatively large size of each membrane-protein system. Nonetheless, the technical approach chosen is based on the methodology that is, in principle, established and widely accessible. Therefore, another group of practitioners would likely be able to reproduce these findings with reasonable effort.

Weaknesses:

The two main results of this paper are (1) that both channels exhibit a flatter structure compared to cryo-EM measurements, and (2) their estimated force vs. displacement relationship. Although the former correlates at least quantitatively with prior experimental work, the latter relies exclusively on simulation results and model parameters.

Below is a summary of the key points we recommend addressing in a revised version of the manuscript:

(1) The authors should report and discuss controls for the membrane energy calculations, specifically by increasing the density of the discretization graining. We also suggest validating the bending modulus used in the energy calculations for the specific lipid mixture employed in the study.

We have addressed both points, see our response to the reviewer's comments for further details.

(2) The authors should consider and discuss the potential limitations of the coarse-grained simulation force field and clarify how atomistic simulations validate the reported results, with a more detailed explanation of the potential interdependencies between the two.

We now discuss the caveats in the comparison of coarse-grained and atomistic simulations in more detail in a new paragraph starting on line 362.

(3) The authors should provide further clarification on other points raised in the reviewers' comments, for instance, the potential role of membrane asymmetry.

We have done this – see above. We now further explain on lines 437ff why we use an asymmetric membrane. On lines 230ff, we discuss that any spontaneous membrane

curvature due to lipid asymmetry is likely small compared to the nanodome curvature and, thus, negligible.

Reviewer #1 (Recommendations for the authors):

(1) Report discretization dependence of the membrane energy (up to double the density of the current discretization graining).

We have added several text pieces in the paragraph “Excess area and bending energy” starting on line 583 in which we state how the results depend on the lattice constant a of the calculations.

(2) Evaluate an analytical energy of a membrane bump with a shape similar to the simulation. This would be free of all sampling and discretization artifacts and would thus be an excellent lower bound of the energy.

We have done this for the curvature profile in Figure 1c and corresponding curvature profiles of the shape profiles in Figure 2d, see next text on lines 326ff.

Minor:

(1) The lipid density (Figure 1 right, 2c, 3c) is not interesting nor is it referred to. It can be dropped.

We think the lipid density maps are important for two reasons: First, they show the protein shape obtained after averaging conformations, as low-lipid-density regions. Second, the lipid densities are used in the calculation of the bending energies, to limit the bending energy calculations to the membrane in the nanodome, see Eq. 9. We therefore prefer to keep them.

(2) Figure 7 is attractive but not used in a meaningful way. I suggest inserting the protein graphic from Figure 7 into Figure 1 with the 4-helix bundles numbered alongside the structure. Figure 7 could then be dropped.

Figure 7 is a figure of the Methods section. We need it to illustrate and explain aspects of the setup (numbering of helices, missing loops) and analysis (numbering scheme of 4-TM helix units).

(3) Some editing of the use of the English language would be helpful. "Exemplary" is a bit of a funny word choice, it implies that the conformation is excellent, and not simply representative. I'd suggest "Representative conformation".

We agree and have replaced “exemplary” by “representative”.

(4) Typos:

Equation 4 - Missing parentheses before squared operator inside the square root.

We have corrected this mistake.

Reviewer #2 (Recommendations for the authors):

This study focuses mainly on Piezo2; the authors do not perform any atomistic simulations of Piezo1, and the coarse-grained simulations for Piezo1 are shorter. As a result, their analysis for Piezo2 seems more complete. It would be good if the authors did similar studies with Piezo1 as with Piezo2.

We agree that atomistic simulations of Piezo 1 would be interesting, too. However, because the atomistic simulations are particularly demanding, this is beyond our scope.

Reviewer #3 (Recommendations for the authors):

(1) At line 63, a very large tension from the previous work by De Vecchis et al is reported (68 mN/m). The authors are sampling values up to about 21 mN/m, which is considerably smaller. However, these values greatly exceed what typical lipid membranes can sustain (about 10 mN/m) before rupturing. When mentioning these large tensions, the authors should emphasize that these values are not physiologically significant, because they would rupture most plasma membranes. That said, their use in simulation could be justified to magnify the structural changes compared to experiments.

We agree that our largest membrane tension values are unphysiological. However, we see a main novelty and relevance of our simulations in the fact that we obtain a response of the nanodome in the physiological range of membrane tensions, see e.g. the 3rd sentence of the abstract. Yes, we include simulations at tensions of 21 mN/m, but most of our simulated tension values are in the range from 0 to 10 mN/m (see e.g. Fig. 3e), in contrast to previous simulation studies.

(2) At line 78 and in the Methods, only the reference paper is for the CHARMM protein force field, but not for the lipid force field.

We have added the reference Klauda et al., 2010 for the CHARMM36 lipid force field in both spots.

(3) (Line 83) Acknowledging that the authors needed to use the structure from micelles (because it has atomic resolution), how closely do their relaxed Piezo structures compare with the lowerresolution data from the MacKinnon and Patapoutian papers?

There are no structures reported in these papers to compare with, only a clear flattening as stated.

(4) (Line 99) The authors chose a slightly asymmetric lipid membrane composition to capture some specific plasma-membrane features. However, they do not discuss which features are described by this particular composition, which doesn't include different acyl-chain unsaturations between leaflets. Further, they do not seem to comment on whether there is enrichment of certain lipid species coupled to curvature, or whether there is any "scrambling" occurring when the dome section and the planar membrane are stitched together in the preparation phase (Figure 8).

Enrichment of lipids in contact with the protein is addressed in the reference Buyan et al., 2020, based on Martini simulations with Piezo 1. We have a different focus, but still wanted to keep an asymmetric membrane as in essentially all previous simulation studies as now stated also on lines 439ff, to mimic the native Piezo membrane environment. There is no apparent "scrambling" in the setup of our membrane systems. We also did not explore any coupling between curvature and lipid composition, but will publish the simulation trajectories to enable such studies.

(5) (Caption of Figure 2). Please comment briefly in the text why the tensionless simulation required a longer simulation run (e.g. larger fluctuations?)

We added as explanation on line 500 as explanation: "... to explore the role of the long-range shape fluctuations in tensionless membranes for the relaxation into equilibrium". The

relaxation time of membrane shape fluctuations strongly increases with the wave length, which is only limited by the simulation box size in the absence of tensions. However, also for 8 microsecond trajectories, we do not observe complete equilibration and therefore decided to extrapolate the excess area and bending energy values obtained for different time intervals of the trajectories.

(6) (Caption of Figure 3). Please clarify in the Methods how the atomistic simulations were initialized were they taken from independent CG simulation snapshots? If not, the use of the adjective "independent" would be questionable given the very short atomistic simulation time length.

We now added that the production simulations started from the same structure. On lines 386, we now discuss the starting structure of the atomistic simulations in more detail.

(7) (Line 202). The approach of discretizing the bilayer shape is reasonable, but no justification was provided for the 1-nm grid spacing. In my opinion, there should be a supporting figure showing how the bending energy varies with the grid spacing.

We now report also the effect of a 2-nm grid spacing on the results, see new text passages on page 18, and provide an explanation for the smaller 1-nm grid spacing on lines 587ff, where we write:

“This lattice constant [$a = 1 \text{ nm}$] is chosen to be smaller than the bin width of about 2nm used in determining the membrane shape of the simulation conformations, to take into account that the averaging of these membrane shapes can lead to a higher resolution compared to the 2 nm resolution of the individual membrane shapes.”

(8) (Line 211). The choice by the authors to use a mixed lipid composition complicates the task of defining a reasonable bending modulus. Experimentally and in atomistic simulations, lipids with one saturated tail (like POPC or SOPC) are much stiffer when they are mixed with cholesterol (<https://doi.org/10.1529/biophysj.105.067652>, <https://doi.org/10.1103/PhysRevE.80.021931>, <https://doi.org/10.1093/pnasnexus/pgad269>). On the other hand, MARTINI seems to predict a slight *softening* for POPC mixed with cholesterol (<https://doi.org/10.1038/s41467-023-43892-x>). Further complicating this matter, mixtures of phospholipids with different preferred curvatures are predicted to be softer than pure bilayers (e.g. <https://doi.org/10.1021/acs.jpcb.3c08117>), but asymmetric bilayers are stiffer than symmetric ones in some circumstances (<https://doi.org/10.1016/j.bpj.2019.11.3398>).

This issue can be quite thorny: therefore, my recommendation would be to either: (a) directly compute k for their lipid composition, which is straightforward when using large CG bilayers (as was done in Fowler et al, 2016), but it would also require more advanced methods for the atomistic ones; (b) use a reasonable *experimental* value for k , based on a similar enough lipid composition.

We now justify in somewhat more detail why we use an asymmetric membrane, but agree that this complicates the bending energy estimates. We only aim to estimate the bending energy in the Martini 2.2 force field, because our elasticity model is based on and, thus, limited to results obtained with this force field. We have included the two further references using the Martini 2.2 force field suggested by the reviewer on line 213, and discuss now in more detail how the bending rigidity estimate enters and affects the modeling, see lines 226ff.

(9) (Line 224). Does this closing statement imply that all experimental work from ex-vivo samples describe Piezo states under some small but measurable tension?

We compare here to the cryo-EM structure in detergent micelles. So, there is no membrane tension, there may be a surface tension of the micelle, but we assume here that Piezo proteins are essentially force free in detergent micelles. Membrane embedding, in contrast, leads to strong forces on Piezo proteins already in the absence of membrane tension, because of the membrane bending energy.

(10) (Line 304). The Discussion concludes with a reasonable point, albeit on a down note: could the authors elaborate on what kind of experimental approach may be able to verify their modeling results?

Very good question, but this is somewhat beyond our expertise. We don't have a clear recommendation – it is complicated. What can be verified is the flattening, i.e. the height and curvature of the nanodome in lower-resolution experiments. We see our results in line with these experiments, see Introduction.

(11) (Line 331). The very title of the Majumder and Straub paper addresses the problem of excessive binding strength between protein beads in the MARTINI force field, which should be mentioned. Figure 3(d) shows that the atomistic systems have larger excess areas than the CG ones. This could be related to MARTINI's "stickiness", or just statistical sampling. Characterizing the grid spacing (see point 7 above) might help illuminate this.

We discuss now the larger excess area values of the atomistic simulations on lines 381ff.

(12) (Lines 367, 375). Are the harmonic restraints absolute position restraints or additional bonds?

Note also that the schedule at which the restraints are released (10-ns intervals) is relatively quick. Does the membrane have enough time to equilibrate the number of lipids in each leaflet?

These are standard, absolute position restraints. The 10-ns intervals may be too short to fully equilibrate the numbers of lipids, we have not explored this. The main point in the setup was to have a reasonable TM helix embedding with a smooth membrane, without any rupturing. This turned out to be tricky, with the procedures illustrated in Figure 8 as solution. If the membrane is smooth, the lipid numbers quickly equilibrate either in the final relaxation or in the initial nanoseconds of the production runs.

(13) (Line 387) The use of an isotropic barostat for equilibration further impedes the system's ability to relax its structure. I feel that the authors should validate more strongly their protocol to rule out the possibility that incomplete equilibration could bias dynamics towards flatter membranes, which is one of the main results of this paper.

We don't see how choices in the initial relaxation steps could have affected our results, at least for the coarse-grained simulations. There is more and more flattening throughout all simulation trajectories, see e.g. the extrapolations in Figure 4. All initial simulation structures are significantly less flattened than the final structures in the production runs.

(14) (Line 403). What is the protocol for reducing the membrane size for atomistic simulation? This is even more important to mention than for CG simulations.

We just cut lipids beyond the intended box size of the atomistic simulations. As a technical point, we now have also added on line 507 how PIP2 lipids were converted.

(15) (Line 423). The CHARMM force field requires a cut-off distance of 12 Å for van der Waals forces, with a force-based continuous switching scheme. The authors should briefly comment on this deviation and its possible impact on membrane properties. Quick test simulations of very small atomistic bilayers with the chosen composition could be used as a comparison.

We don't expect any relevant effect on membrane properties within the statistical accuracies of the quantities of interest here (i.e. excess areas).

(16) (Equation 4). There are some mismatched parentheses: please check.

We have corrected this mistake.

(17) (Equations 7-8). Why did the authors use finite-differences derivatives of $z(x,y)$ instead of using cubic splines and the corresponding analytical derivatives?

In our experience, second derivatives of standard cubic splines can be problematic. The continuous membrane shapes we obtain in our analysis are averages of such splines. We find standard finite differences more reliable, and therefore discretize these shapes. Already for the 2d membrane profiles of Figure 1b and 2d, calculating curvatures from interpolations using splines is problematic.

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